Chapter-1

INTRODUCTION

1.1 Project Background

Modern external beam radiotherapy (EBRT) relies fundamentally on accurate patient anatomical representations to deliver lethal ionizing radiation doses to target neoplastic lesions while strictly sparing adjacent critical organs-at-risk (OARs) [1, 2]. In contemporary clinical oncology, Computed Tomography (CT) serves as the indispensable cornerstone for treatment simulation, anatomical contouring, and radiation dose calculation [2]. The diagnostic values in CT images are natively calibrated in Hounsfield Units (HU), which maintain a direct, monotonic mathematical relationship with physical tissue electron densities (ρ_e) and photon linear attenuation coefficients (μ) [2, 3]:

$$\text{HU} = 1000 \times \left(\frac{\mu_{\text{tissue}} - \mu_{\text{water}}}{\mu_{\text{water}}} \right) \tag{1.1}$$

Because high-energy photon and proton beams interact with human tissue predominantly through Compton scattering and secondary ionization processes, commercial Radiation Treatment Planning Systems (TPS) mandate CT-derived electron density calibration curves to execute Monte Carlo or collapsed-cone convolution dose calculations [2, 4].

Concurrently, Magnetic Resonance Imaging (MRI) has emerged as an indispensable supplementary diagnostic modality in neuro-oncology [3, 4]. By interrogating the nuclear magnetic resonance properties of hydrogen protons under static and radiofrequency magnetic fields, MRI generates exceptional soft-tissue contrast and superior spatial demarcation of intracranial neoplasms, cerebrospinal fluid (CSF) interfaces, and cranial nerve structures that remain indistinct or radiologically isodense on non-contrast CT [1, 4].

However, MRI signal intensity reflects nuclear spin-lattice (T_1) and spin-spin (T_2, T_2^*) relaxation dynamics, proton density, and local magnetic susceptibility gradients rather than physical electron density [2]. Consequently, standard MRI scans lack electron density information and cannot be ingested directly by TPS algorithms for dose deposition modeling [1,2].

1.2 Clinical Need for Synthetic CT (sCT)

Because neither imaging modality alone satisfies both soft-tissue delineation and dose calculation prerequisites, current standard-of-care clinical radiotherapy enforces a dual-modality workflow [2, 4]. In this paradigm, patients must undergo both an initial simulation CT scan (for electron density assignment) and a subsequent high-resolution diagnostic MRI scan (for target volume delineation). Medical physicists then execute multimodal image coregistration to warp the MR-delineated contours onto the reference planning CT geometry [2].

Despite its widespread clinical adoption, this dual-modality acquisition paradigm suffers from three critical vulnerabilities [2–4]:

(1) Systematic Coregistration Errors: Inter-modality image registration between separate CT and MRI scanners inevitably incurs spatial alignment uncertainties ranging from 1.5 mm to over 5.0 mm, even when rigid cranial immobilizers are employed [2, 3]. Anatomical variations—such as patient positioning discrepancies, physiological head flexion, soft-tissue deformation, and non-linear MRI magnetic gradient coil distortions—introduce systematic positioning offsets [2]. In precision stereotactic radiosurgery (SRS) and intensity-modulated proton therapy (IMPT), where treatment margins often do not exceed 1 to 2 mm, registration misalignments can cause critical geometric misses of gross tumor volumes or unacceptable radiation overdose to the brainstem and optic chiasm [4].

(2) Cumulative Ionizing Radiation Exposure: Diagnostic and planning CT scans expose cancer patients to non-negligible ionizing radiation doses (typically 2 to 4 mSv per cranial CT examination) [2]. For pediatric neuro-oncology patients and individuals undergoing repeated adaptive radiotherapy fractions over multi-week protocols, this cumulative secondary radiation significantly elevates long-term risks of radiation-induced secondary malignancies [1, 2].

(3) Elevated Healthcare Costs and Clinical Overhead: The necessity of scheduling, acquiring, reconstructing, and quality-assuring two independent scanning sessions substantially inflates healthcare expenditures, strains hospital infrastructure, and prolongs the crucial time interval between initial cancer diagnosis and radiation treatment delivery [2, 4].

To eliminate these clinical drawbacks, the international radiation oncology community is actively transitioning toward an **"MRI-Only Radiotherapy Planning Workflow"** [1, 2]. In an MRI-only pipeline, the planning CT scan is omitted entirely; instead, a highly accurate *Synthetic CT* (sCT)—also known as a pseudo-CT—is computationally synthesized directly from the planning MRI volume. This sCT preserves the pristine soft-tissue coordinate system of the MR scan while substituting geometrically aligned, quantitatively accurate Hounsfield Units for accurate dosimetric planning [1].

1.3 Technical Challenges in MRI-to-CT Translation

Synthesizing realistic, radiologically valid CT volumes from raw MRI sequences presents severe computational and physical inverse-problem challenges [2, 5, 6]:

- **Severe Signal Ambiguity of Cortical Bone:** The most prominent challenge in MRI-to-CT translation is the acute signal ambiguity between cortical bone and air cavities [1, 2]. Cortical bone possesses an extremely short apparent transverse relaxation time ($T_2^* < 0.5$ ms) and a low mobile proton density; as a result, free induction decay occurs faster than standard MRI echo times (T_E), causing bone to manifest as void, zero-signal hypointense regions in standard T_1-weighted gradient-echo or spin-echo sequences [2]. Similarly, air cavities (such as the sphenoid and ethmoid paranasal sinuses, mastoid air cells, and the ambient scanner bore) contain negligible proton density and appear completely dark [2, 3]. However, on CT, cortical bone and air represent the two opposing extremes of the attenuation spectrum: air measures approximately -1000 HU, whereas dense skull bone exceeds $+1000$ to $+\ 2500$ HU [1]. Differentiating these identically dark MRI regions demands deep spatial contextual reasoning and structural geometric priors.
- **Ill-Posed, Non-Unique Cross-Modality Mapping:** MRI voxel intensities are arbitrary scanner-dependent quantities governed by local coil sensitivities, radiofrequency transmit/receive inhomogeneities (B_1^+ and B_1^- fields), receiver gain, and vendor-specific reconstruction filters [1]. Consequently, MRI voxel values lack absolute physical units, rendering the translation f : MRI → CT an intrinsically non-injective, ill-posed inverse problem [5, 6].
- **Geometric and Susceptibility Distortions:** Static magnetic field (B_0) inhomogeneities at air-tissue interfaces cause localized voxel displacement and distortion artifacts in MRI,

which contrast sharply with the rigid, geometrically faithful reconstruction geometry of CT [2].

1.4 Problem Statement and Mathematical Formulations

Given an input planning magnetic resonance image slice $\mathbf{x} \in \mathrm{R}^{H \times W}$ within a spatial domain $\Omega \subset \mathrm{R}^2$, the objective of MRI-to-CT image translation is to learn a parameterized functional mapping $f_\vartheta : \mathbf{x} \to \hat{\mathbf{y}}$ that accurately estimates the true, co-registered computed tomography slice $\mathbf{y} \in \mathrm{R}^{H \times W}$ across all anatomical structures. In accordance with clinical benchmarks established by the international SynthRAD2023 challenge [1], both MRI and CT volumes are scaled to a unified, normalized physical dynamic range $[-1.0, +1.0]$, corresponding to the standard clinical CT window [−1000 HU, +3000 HU] via the linear transformation:

$$y_{\text{norm}} = 2 \times \frac{(y_{\text{HU}} - \text{HU}_{\min})}{\text{HU}_{\max} - \text{HU}_{\min}} - 1.0, \qquad \text{where HU}_{\min} = -1000,\ \text{HU}_{\max} = 3000 \quad (1.2)$$

To thoroughly investigate generative deep learning paradigms, this project formulates, implements, and benchmarks five distinct model architectures under strictly identical training, validation, and testing splits (Seed 42, 80%/20% patient-level cohort split):

1.4.1 Model 1: Deep Residual U-Net (ResUNet Baseline)

The first model implements a supervised, fully convolutional baseline utilizing deep residual blocks to mitigate vanishing gradients and facilitate multi-scale feature propagation [7,8]. The network parameters $\vartheta_{\text{ResUNet}}$ are optimized using a composite objective balancing voxel-wise fidelity (L_1) and global reconstruction energy (Mean Squared Error, MSE):

$$\mathrm{L}_{\text{ResUNet}}(\vartheta) = \frac{1}{|\Omega|} \sum_{i \in \Omega} |\hat{y}_i - y_i| + \frac{\lambda_{\text{mse}}}{|\Omega|} \sum_{i \in \Omega} (\hat{y}_i - y_i)^2 \quad (1.3)$$

While computationally efficient and deterministic, pure regression models notoriously suffer from regression-to-the-mean effects, leading to oversmoothed bone-tissue boundaries and structural attenuation underestimations.

1.4.2 Model 2: Conditional Generative Adversarial Network (Pix2Pix cGAN)

To overcome blurriness and encourage realistic high-frequency anatomical texture, Model 2 implements a conditional adversarial framework [9, 10]. The generator G and a Markovian PatchGAN discriminator D engage in a two-player minimax game conditioned on the MRI image $\mathbf{x}$:

$$\min_G \max_D \mathrm{L_{cGAN}}(G, D) + \lambda_{\mathrm{rec}} \mathrm{L}_{L_1}(G) \tag{1.4}$$

where the conditional adversarial objective is defined as:

$$\mathrm{L_{cGAN}}(G, D) = \mathrm{E}_{\mathbf{x},\mathbf{y}} [\log D(\mathbf{x}, \mathbf{y})] + \mathrm{E}_{\mathbf{x}} [\log (1 - D(\mathbf{x}, G(\mathbf{x})))] \tag{1.5}$$

and $\mathrm{L}_{L_1}(G) = \mathrm{E}_{\mathbf{x},\mathbf{y}} [\| \mathbf{y} - G(\mathbf{x}) \|_1]$. Although Pix2Pix produces visually sharper bone structures, adversarial training is susceptible to mode collapse, hallucinated pseudo-anatomical structures, and subtle geometric distortions unacceptable for radiation dosimetry.

1.4.3 Model 3: Conditional Denoising Diffusion Probabilistic Model (cDDPM From Scratch)

Model 3 explores likelihood-based generative diffusion modeling [11, 12]. Unlike GANs, diffusion models define a rigorous forward Gaussian corruption process $q(\mathbf{y}_t|\mathbf{y}_{t-1})$ parameterized by a predetermined variance schedule $\{\beta_t \in (0, 1)\}_{t=1}^{T}$:

$$q(\mathbf{y}_t|\mathbf{y}_0) = \mathrm{N}\left(\mathbf{y}_t; \sqrt{\bar{\alpha}_t}\mathbf{y}_0, (1 - \bar{\alpha}_t)\mathbf{I}\right), \qquad \alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s \tag{1.6}$$

The neural denoiser $\boldsymbol{\epsilon}_\vartheta(\mathbf{y}_t, t, \mathbf{x}, \mathbf{m})$ receives a 3-channel input tensor concatenating the noisy intermediate CT $\mathbf{y}_t$, conditioning MRI $\mathbf{x}$, and brain mask $\mathbf{m}$ to predict the injected Gaussian noise $\boldsymbol{\epsilon} \sim \mathrm{N}(\mathbf{0}, \mathbf{I})$:

$$\mathrm{L_{cDDPM}}(\vartheta) = \mathrm{E}_{t,\mathbf{y}_0,\boldsymbol{\epsilon},\mathbf{x},\mathbf{m}} \left[\left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\vartheta \left(\sqrt{\bar{\alpha}}\mathbf{y}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}, t, \mathbf{x}, \mathbf{m} \right) \right\|_2^2 \right] \tag{1.7}$$

Inference is accelerated using Denoising Diffusion Implicit Models (DDIM, $\eta = 0$) across 50 deterministic reverse steps [13].

1.4.4 Model 4: SOTA Conditional DDPM with Residual Attention (SOTA-CDDPM)

Model 4 extends diffusion modeling by incorporating cross-layer Attention Gates and an anatomical multi-component hybrid loss formulation [5, 6]. Attention gates dynamically suppress irrelevant background scanner noise and highlight steep cranial gradient interfaces:

$$\alpha = \sigma\left(\psi^{T}\left[\mathrm{ReLU}\left(W_g\mathbf{g} + W_x\mathbf{x}\right)\right]\right) \tag{1.8}$$

The training objective couples standard noise prediction with bone-weighted absolute error and structural edge preservation:

$$L_{SOTA} = L_{diff}(\boldsymbol{\epsilon}, \hat{\boldsymbol{\epsilon}}) + \lambda_{bone}\mathrm{E}\left[\mathbf{w}_{bone} \odot |\hat{\mathbf{y}}_0 - \mathbf{y}_0|\right] + \lambda_{ssim}L_{SSIM}(\hat{\mathbf{y}}_0, \mathbf{y}_0) \tag{1.9}$$

where $\mathbf{w}_{bone} = 1.0 + 4.0 \cdot \mathrm{I}(y_{real} > 300\ \mathrm{HU})$ quadruples the penalty on erroneous skull predictions.

1.4.5 Model 5: Proposed Cross-Conditioned Residual Diffusion (CCR-Diffusion)

To overcome the high sampling latency and residual noise artifacts of standard diffusion models, Model 5 establishes a novel two-stage decoupled paradigm:

1. **Stage 1 (Coarse Anatomical Anchor):** A 4-stage residual network generates a deterministic structural coarse CT $\mathbf{c} = f_{stage1}(\mathbf{x}) \in [-1.0, 1.0]$. Multi-scale feature tensors $\mathbf{F}^{(l)}_{enc} \in \mathrm{R}^{C_l \times H_l \times W_l}$ are extracted across all encoder depths ($l \in \{1, 2, 3, 4\}$).
2. **Stage 2 (FiLM-Modulated Residual Diffusion):** The diffusion model operates exclusively on normalized high-frequency residual vectors $\mathbf{z}_0 = (\mathbf{y} - \mathbf{c})/s$, where $s = \sigma(\mathbf{y} - \mathbf{c})$ is the empirical residual scale. Decoder feature representations are dynamically modulated via Feature-wise Linear Modulation (FiLM) [14]:

$$\mathrm{FiLM}(\mathbf{F}; \boldsymbol{\gamma}, \boldsymbol{\beta}) = (1.0 + \boldsymbol{\gamma}) \odot \mathbf{F} + \boldsymbol{\beta} \tag{1.10}$$

where $\boldsymbol{\gamma}, \boldsymbol{\beta} = \mathrm{Conv}_{1\times1}(\mathbf{F}^{(l)}_{enc})$. Reconstructed synthetic CT is obtained deterministically

via:

$$\hat{\mathbf{y}} = \text{clip}\,(\mathbf{c} + s \cdot \mathbf{z}_0,\ -1.0,\ 1.0) \tag{1.11}$$

1.5 Project Objectives

The central goal of this project is to develop, benchmark, and clinically validate an end-to-end deep learning framework for MRI-to-CT translation capable of enabling MRI-only radiotherapy planning. The specific measurable objectives are:

1. **Standardized Preprocessing and Cohort Stratification:** Construct a robust, reproducible data processing pipeline adhering to the SynthRAD2023 challenge protocol, incorporating robust intensity normalization ([−1000 HU, +3000 HU] → [−1.0, +1.0]), cranial mask isolation, and rigorous patient-level 80%/20% split isolation (Seed 42) to prevent data leakage.
2. **Rigorous 5-Model Comparative Implementation:** Implement, train to convergence, and evaluate five diverse architectural families spanning deep convolutional regression (ResUNet), generative adversarial networks (Pix2Pix cGAN), unconstrained conditional diffusion (cDDPM), attention-guided diffusion (SOTA-CDDPM), and the proposed two-stage Cross-Conditioned Residual Diffusion (CCR-Diffusion).
3. **Multi-Faceted Clinical Metric Benchmarking:** Evaluate synthesized CT slices across four primary quantitative metrics: Mean Absolute Error across all intracranial tissues (MAE_{all}), Bone-specific reconstruction error (MAE_{bone}, HU > 300), Structural Similarity Index Measure (SSIM), and Peak Signal-to-Noise Ratio (PSNR), aggregated across both slice-level and patient-level distributions.
4. **Computational and Hardware Profiling:** Measure real-world clinical feasibility by profiling each model's total parameter footprint, trainable parameters, single-slice inference latency (milliseconds per slice), frames per second (FPS), and peak GPU VRAM utilization under deterministic DDIM sampling (50 steps).
5. **Ablation and Radiotherapy Dosimetric Validation:** Validate the superiority of two-stage residual diffusion over standard GAN and DDPM models, demonstrating significant reductions in cortical bone error (MAE_{bone}) and eliminating hallucinated anatomical artifacts.

1.6 Scope and Boundaries

The scope of this major project encompasses:

- **Anatomical Site:** Brain cancer radiotherapy imaging from the SynthRAD2023 Grand Challenge dataset (comprising 363 paired 2D axial test slices across diverse patient demographics).
- **Input Modality:** Paired T_1-weighted axial Magnetic Resonance Imaging (MRI) rigidly registered to planning CT volumes.
- **Output Modality:** Quantitative Synthetic Computed Tomography (sCT) calibrated in absolute Hounsfield Units within the $[-1000$ HU, $+3000$ HU] window.
- **Hardware Scope:** Model training and GPU accelerated inference benchmarking conducted on NVIDIA GPU infrastructure using PyTorch and CUDA runtime environments.

Project Boundaries: The project specifically focuses on 2D axial slice-wise generation. Inter-fraction anatomical changes occurring during multi-week radiotherapy (e.g., tumor shrinkage or weight loss) and non-rigid deformable organ motion outside the intracranial cavity (such as thoracic or abdominal respiratory motion) lie beyond the immediate scope of this study.

1.7 Organization of the Report

The remainder of this major project report is structured into five subsequent chapters:

- **Chapter 2 (Literature Survey):** Presents an exhaustive review of related work, tracing the historical evolution of synthetic CT generation from atlas-based and bulk-density assignment methods to modern Convolutional Neural Networks, Generative Adversarial Networks, and Denoising Diffusion Probabilistic Models. A comprehensive comparison table highlights gaps in existing literature.
- **Chapter 3 (System Requirements and Methodology):** Formulates the end-to-end system architecture, mathematical principles of the five generative pipelines, dataset specifications, preprocessing pipelines, and Data Flow Diagrams (DFDs).
- **Chapter 4 (Implementation Details):** Details the software engineering stack, modular codebase structure, hyperparameter configurations, loss function implementations, and DDIM reverse sampling algorithms.

- **Chapter 5 (Results, Quantitative Evaluation, and Discussion):** Provides exhaustive empirical results, including comparative metric tables (MAE_{all}, MAE_{bone}, SSIM, PSNR), hardware latency and VRAM profiling, qualitative visual comparison grids, error difference maps, and clinical dosimetric impact analyses.
- **Chapter 6 (Conclusion and Future Scope):** Summarizes the major research findings, outlines clinical translation feasibility, acknowledges project limitations, and outlines future avenues for 3D volumetric extension and adaptive proton therapy integration.

Chapter-2

LITERATURE SURVEY

2.1 Introduction to the Domain Literature

The quest to achieve an accurate, robust, and clinically deployable MRI-only radiotherapy workflow has inspired intensive multidisciplinary research spanning medical physics, clinical oncology, computer vision, and deep generative learning over the past two decades [2]. The central objective of MRI-to-CT translation is to infer quantitatively reliable Hounsfield Units (HU) and spatial electron density distributions (ρ_e) directly from non-ionizing magnetic resonance acquisitions, thereby resolving the dual-modality registration dilemmas inherent to modern treatment planning [1, 2].

Methodologies developed in the literature can be broadly categorized into five successive evolutionary generations:

(1) Classical Heuristic Methods (Pre-Deep Learning): Including bulk-density assignment through multi-tissue thresholding, fuzzy classification, and single/multi-atlas deformable image registration [15–20].

(2) Deep Convolutional Regression Architectures: Employing supervised voxel-wise loss formulations (L_1, L_2, Huber) over fully convolutional architectures, such as standard U-Nets, 3D CNNs, and Residual U-Nets [7, 8, 21–26].

(3) Generative Adversarial Networks (GANs): Leveraging minimax game dynamics with PatchGAN discriminators to synthesize high-frequency structural textures in paired (Pix2Pix) and unpaired (CycleGAN, MedGAN) imaging regimes [9, 10, 27–32].

(4) Denoising Diffusion Probabilistic Models (DDPMs): Establishing non-adversarial, likelihood-based generative formulations capable of modeling complex anatomical probability distributions via iterative stochastic denoising and deterministic DDIM trajectories [6, 11–13, 33–36].

(5) Hybrid, Residual, and Feature-Modulated Generative Models: Decoupling global structural prediction from high-frequency residual diffusion using Feature-wise Linear Modulation (FiLM) and latent representations [1, 5, 14, 37, 38].

This chapter conducts a comprehensive critical examination of each methodological paradigm, identifying their architectural foundations, clinical capabilities, and persistent technical bottlenecks across 35 landmark studies.

2.2 Classical Atlas-Based and Bulk-Density Assignment Approaches

2.2.1 Bulk Tissue Density Assignment (HU Override)

Early investigations in MRI-based radiotherapy relied on direct tissue segmentation combined with bulk Hounsfield Unit overriding [2, 15]. In this heuristic framework, the patient's planning MRI is segmented into a small number of discrete tissue classes (typically 3 to 5 compartments: air, lung, soft tissue, spongy bone, and dense cortical bone) using intensity thresholding, fuzzy c-means clustering, or dual-echo ultrashort echo time (UTE) sequences [15, 18]. Each segmented voxel is subsequently assigned a uniform, literature-standard CT number (for example, −1000 HU for air, +40 HU for cranial soft tissue, and +1000 HU for cortical bone).

While computationally trivial and conceptually straightforward, bulk-density assignment models suffer from fundamental physical approximations [2]:

- **Neglect of Intra-Tissue Density Heterogeneity:** Real human bone possesses complex trabecular architecture, diploic spaces, and varying mineralization gradients. Enforcing a homogeneous HU override obliterates intra-organ density variations, leading to systematic dose calculation discrepancies of 2% to 5% in photon therapy and exceeding 10% in proton Bragg peak range estimations [2, 4].
- **Segmentation Boundary Sensitivity:** Any misclassification at tissue interfaces (such as classifying thin skull bone as air due to partial volume effects) introduces severe step-function density errors that propagate directly into radiation dose calculations [3].

2.2.2 Single-Atlas and Multi-Atlas Deformable Registration

To capture continuous spatial attenuation gradients, atlas-based methods were subsequently introduced [16,17,19,20]. In single-atlas approaches, a pre-acquired exemplar patient dataset containing perfectly co-registered MRI and CT volumes is used as a template. Non-rigid

deformable image registration (DIR) algorithms warp the atlas MRI to match the target patient's MRI anatomy; the resulting coordinate deformation field $\boldsymbol{\phi}(\mathbf{x})$ is then applied directly to the atlas CT to construct the patient-specific synthetic CT [16]:

$$\hat{\mathbf{y}}_{\text{target}}(\mathbf{x}) = \mathbf{y}_{\text{atlas}}(\boldsymbol{\phi}(\mathbf{x})) \tag{2.1}$$

To mitigate single-atlas bias and morphological idiosyncrasies, multi-atlas frameworks compile databases of N diverse patient templates ($N \in [10, 30]$) [16, 19]. Target MRIs are registered independently to all N atlas MRIs, and the warped candidate CT volumes are fused using locally weighted intensity voting or patch-based similarity matching:

$$\hat{\mathbf{y}}(\mathbf{x}) = \sum_{k=1}^{N} w_k(\mathbf{x}) \cdot \mathbf{y}_k(\boldsymbol{\phi}_k(\mathbf{x})), \quad \sum_{k=1}^{N} w_k(\mathbf{x}) = 1 \tag{2.2}$$

where $w_k(\mathbf{x})$ reflects local patch correlation in MRI space.

2.2.3 Fundamental Limitations of Classical Approaches

Despite delivering continuous attenuation distributions superior to bulk overrides, atlas-based methods exhibit critical clinical shortcomings [1, 2]:

(1) Inability to Model Atypical and Pathological Anatomies: Deformable registration assumes topological homeomorphism between atlas and patient. In patients presenting with massive intracranial tumors, midline shifts, craniotomy defects, or post-surgical resection cavities, non-rigid registration inevitably fails, producing distorted anatomical hallucinations [2].

(2) Extreme Computational Latency: Multi-atlas non-rigid registrations across large 3D patient volumes require substantial iterative optimization, routinely consuming 30 minutes to 2 hours per patient volume, which is incompatible with online adaptive radiotherapy workflows [16].

2.3 Deep Convolutional Regression Networks

2.3.1 Fully Convolutional Networks and U-Net Architectures

The advent of deep convolutional neural networks revolutionized medical image-to-image translation [8, 21, 22]. Han (2017) demonstrated the first end-to-end deep learning framework for cranial synthetic CT generation using a deep convolutional autoencoder architecture [21]. By learning hierarchical spatial feature representations directly from paired MRI and CT voxels, the network mapped complex non-linear intensity relationships without explicit tissue segmentation.

The quintessential architecture for biomedical translation is the U-Net [8], comprising an encoder contracting path that captures contextual semantic abstraction, a symmetric decoder expanding path that enables precise spatial localization, and direct skip connections between opposing hierarchical levels. Skip connections concatenate high-resolution feature maps from encoder stages directly to decoder upsampling stages, preserving fine spatial edge boundaries that would otherwise be lost through pooling or strided convolutions [8, 24].

2.3.2 Deep Residual U-Net (ResUNet) and Gradient Dynamics

Standard U-Nets encounter optimization difficulties when scaled to very deep topologies due to vanishing and exploding gradient phenomena [39]. To stabilize optimization and accelerate feature propagation across deep receptive fields, Zhang et al. [7] and Fu et al. [25] formulated the Deep Residual U-Net (ResUNet). Within each convolutional block, identity shortcut connections are introduced:

$$\mathbf{h}_l = \sigma\left(\mathbf{h}_{l-1} + \mathrm{F}(\mathbf{h}_{l-1}, \mathrm{W}_l)\right) \tag{2.3}$$

where $\mathrm{F}(\cdot)$ represents stacked 3×3 convolutions, batch/instance normalization, and non-linear activations $\sigma(\cdot)$. Residual connections ensure uninterrupted backpropagation of error gradients:

$$\frac{\partial \mathrm{E}}{\partial \mathbf{h}_{l-1}} = \frac{\partial \mathrm{E}}{\partial \mathbf{h}_l}\left(1 + \frac{\partial \mathrm{F}}{\partial \mathbf{h}_{l-1}}\right) \tag{2.4}$$

thereby enabling stable training of high-capacity models without degradation.

2.3.3 The Regression-to-the-Mean Phenomenon and High-Frequency Loss

Despite yielding high Peak Signal-to-Noise Ratio (PSNR) values and low average Mean Absolute Error (MAE), direct convolutional regression networks suffer from a well-known mathematical artifact termed *regression-to-the-mean* [9, 12]. When optimized using pixel-wise L_1 or L_2 objectives:

$$\vartheta^* = \arg\min_{\vartheta} \mathrm{E}_{\mathbf{x},\mathbf{y}}\left[\|\mathbf{y} - f_\vartheta(\mathbf{x})\|^p\right], \quad p \in \{1, 2\} \tag{2.5}$$

the network estimates the conditional expectation $\mathrm{E}[\mathbf{y}|\mathbf{x}]$ (for $p = 2$) or conditional median (for $p = 1$) across all plausible target textures consistent with input $\mathbf{x}$ [9]. In areas of high structural ambiguity—such as complex cranial sutures, mastoid air cell honeycombs, and inner ear labyrinths—the model averages multiple conflicting high-frequency configurations. This mathematical averaging manifests as severe structural oversmoothing and attenuation underestimation in cortical bone, which directly perturbs dose calculations at bone-soft tissue interfaces [2].

2.4 Generative Adversarial Networks for Cross-Modality Synthesis

2.4.1 Conditional GANs (Pix2Pix) and PatchGAN Discriminators

To force neural generators to synthesize crisp high-frequency textures rather than smooth ensemble averages, Isola et al. (2017) introduced the conditional Generative Adversarial Network (Pix2Pix) [9]. Adapted to medical imaging by Nie et al. [10] and Maspero et al. [29], Pix2Pix frames MRI-to-CT translation as an adversarial minimax game between a generator G_ϑ and an auxiliary discriminator D_ϕ:

$$\min_G \max_D \mathrm{L}_{\mathrm{cGAN}}(G, D) + \lambda \mathrm{L}_{L_1}(G) \tag{2.6}$$

where:

$$\mathrm{L}_{\mathrm{cGAN}}(G, D) = \mathrm{E}_{\mathbf{x},\mathbf{y}}[\log D(\mathbf{x}, \mathbf{y})] + \mathrm{E}_{\mathbf{x}}[\log(1 - D(\mathbf{x}, G(\mathbf{x})))] \tag{2.7}$$

A pivotal innovation of Pix2Pix is the *Markovian PatchGAN* discriminator [9]. Rather than evaluating the entire image as a single global real/fake scalar, the PatchGAN restricts its receptive field to local $N \times N$ image patches (e.g., 70×70 voxels), classifying whether each sub-patch represents a realistic cross-modality pairing. By penalizing local structural incongruities, the PatchGAN functions as a learned high-frequency texture loss, compelling G_ϑ to recover realistic bone edges and cranial boundaries [10, 29, 31].

2.4.2 Unpaired Cross-Modality Translation (CycleGAN)

While Pix2Pix mandates strictly registered paired training slices, acquiring geometrically congruent MRI-CT datasets is challenging in pelvic and thoracic sites due to physiological peristalsis, rectal filling, and respiratory deformation [27]. To circumvent rigid registration requirements, Wolterink et al. [27] and Zhu et al. [28] introduced Cycle-Consistent GANs (CycleGAN). Operating on unpaired datasets X ⊂ MRI and Y ⊂ CT, CycleGAN couples dual generator-discriminator pairs with a cycle-consistency constraint:

$$\mathrm{L}_{\mathrm{cyc}}(G_{X\to Y}, F_{Y\to X}) = \mathrm{E}_{\mathbf{x}}\left[\| F(G(\mathbf{x})) - \mathbf{x}\|_1\right] + \mathrm{E}_{\mathbf{y}}\left[\| G(F(\mathbf{y})) - \mathbf{y}\|_1\right] \qquad (2.8)$$

ensuring that forward translation followed by inverse synthesis recovers the original input volume.

2.4.3 Pathological Hallucinations and Dosimetric Vulnerabilities of GANs

Although GAN-based sCT models produce visually compelling images with sharp edges, their deployment in clinical radiation oncology remains intensely scrutinized due to grave safety concerns [1, 2, 5]:

- **Adversarial Hallucination Artifacts:** The adversarial objective optimizes perceptual plausibility rather than physical photon attenuation truth [5]. Consequently, GAN generators are prone to hallucinating fictitious bone calcifications, missing pathological tumor lesions, or fabricating non-existent air pockets to satisfy the discriminator's local texture criteria [1, 32].
- **Mode Collapse and Training Instability:** Adversarial optimization represents an unstable saddle-point search prone to limit cycles, mode collapse, and hyperparameter sensitiv-

ity [33]. In clinical regimes, such unpredictability can lead to catastrophic synthesis failures on out-of-distribution patient scans.

2.5 Denoising Diffusion Probabilistic Models in Medical Imaging

2.5.1 Foundations of Forward and Reverse Gaussian Processes

Denoising Diffusion Probabilistic Models (DDPMs) have recently emerged as the preeminent generative modeling paradigm, systematically outperforming GANs in image quality, distribution coverage, and training stability [11, 33]. Grounded in non-equilibrium thermodynamics, DDPMs model the target data distribution $p(\mathbf{y}_0)$ through two dual Markov chains [11]:

1. Forward Corruption Process: Given a ground-truth CT slice $\mathbf{y}_0$, the forward diffusion process progressively adds isotropic Gaussian noise across T discrete timesteps ($t \in \{1, \ldots, T\}$) according to a predefined variance schedule $\{\beta_t\}_{t=1}^{T}$:

$$q(\mathbf{y}_t|\mathbf{y}_{t-1}) = \mathrm{N}\left(\mathbf{y}_t;\ \sqrt{1-\beta_t}\mathbf{y}_{t-1},\ \beta_t\mathbf{I}\right) \tag{2.9}$$

Using the property of Gaussian marginalization, any intermediate noisy state $\mathbf{y}_t$ can be sampled in closed form directly from $\mathbf{y}_0$:

$$\mathbf{y}_t = \sqrt{\bar{\alpha}_t}\mathbf{y}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathrm{N}(\mathbf{0}, \mathbf{I}) \tag{2.10}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. As $t \to T$, $\bar{\alpha}_T \to 0$, causing $\mathbf{y}_T$ to converge to pure spherical Gaussian noise $\mathrm{N}(\mathbf{0}, \mathbf{I})$.

2. Reverse Denoising Process: The reverse process seeks to reconstruct the clean image $\mathbf{y}_0$ by inverting the noise corruption trajectory:

$$p_\vartheta(\mathbf{y}_{t-1}|\mathbf{y}_t) = \mathrm{N}\left(\mathbf{y}_{t-1};\ \boldsymbol{\mu}_\vartheta(\mathbf{y}_t, t),\ \boldsymbol{\Sigma}_\vartheta(\mathbf{y}_t, t)\right) \tag{2.11}$$

Ho et al. [11] demonstrated that parameterizing the mean $\boldsymbol{\mu}_\vartheta$ as a noise predictor $\boldsymbol{\epsilon}_\vartheta(\mathbf{y}_t, t)$ produces superior empirical sample quality under the simplified L_2 objective:

$$L_{\text{simple}}(\vartheta) = \mathbb{E}_{t,\mathbf{y}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\vartheta(\mathbf{y}_t, t)\right\|^2\right] \tag{2.12}$$

2.5.2 Conditioning Paradigms for Medical Image Translation

To repurpose unconditional diffusion for cross-modality medical translation (f : MRI → CT), the reverse trajectory must be steered by the conditioning MRI image $\mathbf{x}$ [6, 12]. Saharia et al. (Palette) [12] established channel-wise input concatenation:

$$\mathbf{h}_{\text{in}} = [\mathbf{y}_t \,\|\, \mathbf{x}] \in \mathbb{R}^{(C_y+C_x)\times H\times W} \tag{2.13}$$

feeding both the corrupted target and pristine condition into the neural denoiser $\boldsymbol{\epsilon}_\vartheta(\mathbf{y}_t, t, \mathbf{x})$. Lyu and Wang [6], Gong et al. [34], and Dorjsembe et al. [35] applied conditional diffusion to MRI-CT translation, demonstrating that diffusion models entirely eliminate GAN hallucination artifacts while preserving high-contrast skull boundaries.

2.5.3 Sampling Acceleration (DDIM) and Computational Latency Bottlenecks

The critical impediment to clinical adoption of standard DDPMs is sampling latency [13]. Generating a single synthetic CT slice via ancestral Langevin sampling requires iterating through $T = 1000$ sequential forward neural evaluations, demanding 20 to 60 seconds per 2D slice on clinical GPUs [5, 13].

Song et al. [13] formulated Denoising Diffusion Implicit Models (DDIM), establishing a non-Markovian deterministic sampling trajectory that permits skipping timesteps along a sub-sequence $\tau = [\tau_1, \tau_2, \ldots, \tau_S]$ with $S \ll T$:

$$\mathbf{y}_{\tau_{i-1}} = \sqrt{\bar{\alpha}_{\tau_{i-1}}}\left(\frac{\mathbf{y}_{\tau_i} - \sqrt{1-\bar{\alpha}_{\tau_i}}\,\boldsymbol{\epsilon}_\vartheta(\mathbf{y}_{\tau_i}, \tau_i, \mathbf{x})}{\sqrt{\bar{\alpha}_{\tau_i}}}\right) + \sqrt{1-\bar{\alpha}_{\tau_{i-1}}}\,\boldsymbol{\epsilon}_\vartheta(\mathbf{y}_{\tau_i}, \tau_i, \mathbf{x}) \tag{2.14}$$

DDIM reduces reverse sampling to $S = 50$ steps, accelerating inference by an order of magnitude. Nonetheless, when synthesizing an entire 3D patient volume containing 150 to 300 slices, pure diffusion still incurs 3 to 8 minutes of GPU computation, motivating

architectural innovations in residual diffusion.

2.6 Feature Conditioning and Residual Diffusion Mechanisms

2.6.1 Decoupled Coarse-to-Fine Architectures

To alleviate the representational burden on diffusion models, recent investigations explore decoupled coarse-to-fine paradigms [5, 14, 38]. In naive conditional DDPM, the network must simultaneously perform two disparate tasks:

1. Learning global anatomical layout and soft-tissue organ boundaries from scratch.
2. Synthesizing fine-grained Hounsfield Unit variations and cortical bone microstructures.

By decoupling these responsibilities into a deterministic Stage-1 predictor and a generative Stage-2 residual refiner, the diffusion model is relieved of reconstructing macro-scale anatomy, concentrating its modeling capacity solely on localized high-frequency residuals [14].

2.6.2 Feature-Wise Linear Modulation (FiLM)

Integrating conditioning information solely via early channel concatenation (as in Palette) often leads to conditional neglect, where the deep layers of the denoiser lose alignment with the conditioning MRI features [14]. Feature-wise Linear Modulation (FiLM) [14] overcomes this by applying affine transformations to intermediate activation maps throughout the network:

$$\mathrm{FiLM}(\mathbf{F}; \boldsymbol{\gamma}, \boldsymbol{\beta}) = (1.0 + \boldsymbol{\gamma}(\mathbf{c})) \odot \mathbf{F} + \boldsymbol{\beta}(\mathbf{c}) \tag{2.15}$$

where γ and β are adaptive scaling and shifting vectors dynamically predicted from the conditioning structural features c via 1 × 1 convolutional projections. FiLM ensures that anatomical geometry actively gates and modulates feature representations at every hierarchical depth of the generative denoiser. Unlike naive input concatenation—which treats spatial anatomical priors as passive early-stage channels—multi-scale FiLM layers inject hierarchical conditioning directly into the bottleneck and decoding blocks at resolutions of 16×16, 32×32, and 64×64. By adaptively scaling and shifting normalized feature activations, the network dynamically amplifies high-frequency calvarial bone textures while suppressing stochastic variance across homogeneous brain parenchyma. Consequently, the diffusion trajectory is strictly bounded by the macroscopic coordinates

established in Stage 1, eliminating geometric drift and ensuring that stochastic reverse sampling refines only true physical Hounsfield Unit variations without compromising global anatomical integrity.

2.7 Comprehensive Literature Survey Summary Matrix

To systematically analyze the trajectory of MRI-to-CT translation research and situate our five evaluated architectures, Tables 2.1 through 2.5 present a comprehensive comparative matrix across 35 landmark studies published between 2013 and 2026. These investigations are categorized chronologically across atlas-based mapping, supervised deep convolutional regression, adversarial synthesis, and modern score-based diffusion paradigms. Each taxonomy outlines anatomical target sites, loss formulations, reported quantitative metrics, and clinical limitations, contextualizing the technical evolution toward residual diffusion modeling.

2.7.1 Category 1: Classical Bulk Assignment and Multi-Atlas DIR Methods

Table 2.1: Literature Survey: Classical Bulk Density Overrides and Multi-Atlas DIR

Sl.	Author & Year	Architecture	Dataset	Performance	Critical Research Gaps
1	Kapanen & Tenhunen (2013) [15]	Dual-echo bulk HU override	Cranial T_1/T_2^*	Soft dose diff < 2%	Assumes bone homogeneity; large step errors at cranial boundaries.
2	Sjölund et al. (2015) [16]	Multi-Atlas non-rigid DIR	Cranial T_1	MAE ≈ 65 HU	High runtime (30–60 min); cannot adapt to craniotomies or tumors.
3	Gudur et al. (2014) [17]	Atlas deformable registration	Head and Neck	MAE = 78 HU	Sensitive to non-rigid swallowing artifacts and atlas template bias.
4	Korhonen et al. (2014) [18]	Dual-model bulk segmentation	Brain (20 pts)	Dose dev < 1.5%	Ignores micro-trabecular bone structures and diploic space variation.
5	Burgos et al. (2014) [19]	Multi-atlas patch fusion	Brain PET/MR	MAE = 72 HU	High computational overhead; fails under severe pathological shift.
6	Dowling et al. (2015) [20]	Pelvis (40 pts)	MAE ≈ 58 HU	Prone to registration failure caused by	gular gas bowel pockets.
`	Atlas-based regression			i r r e	

2.7.2 Category 2: Deep Convolutional Neural Networks and Regression Baselines

Table 2.2: Literature Survey: Deep Convolutional Neural Networks and Regression Baselines

Sl.	Author & Year	Architecture	Dataset	Performance	Critical Research Gaps
7	Han (2017) [21]	Deep U-Net (L_1 loss)	Brain (18 pts)	MAE = 84.8 HU, PSNR = 27.2 dB	Severe regression-to-mean causes over-smoothed skull edges and low peak HU.
8	Nie et al. (2016) [22]	Fully Convolutional FCN	Brain MRI	MAE ≈ 90 HU	Coarse spatial outputs; lacks multi-scale context needed for bone density.
9	Zhang et al. (2018) [7]	Deep Residual U-Net	Remote / Medical	Improved stability	Identity skips improve convergence but pixel loss remains blurry.
10	Jin et al. (2019) [23]	3D Deep CNN	Pelvis (64 pts)	MAE = 51.2 HU	Excessive GPU VRAM demands in 3D; smooths thin bony cortical rims.
11	Dinkla et al. (2018) [24]	Dilated Convolutional Net	Brain (46 pts)	MAE = 65 HU	Dilated kernels produce gridding artifacts in skull bone synthesis.
12	Fu et al. (2019) [25]	ResUNet with Huber loss	Head/Neck	MAE = 60.5 HU	Huber thresholding reduces outliers but bone error remains high (> 200 HU).
13	Arabi et al. (2018) [26]	Deep autoencoder ensemble	Whole-body MR	MAE ≈ 76 HU	Inter-slice discontinuities and intensity drift across coil boundaries.

2.7.3 Category 3: Generative Adversarial Networks (GANs)

Table 2.3: Literature Survey: Generative Adversarial Networks (Pix2Pix, CycleGAN, MedGAN)

Sl.	Author & Year	Architecture	Dataset	Performance	Critical Research Gaps
14	Isola et al. (2017) [9]	Pix2Pix Conditional GAN	Vision / Paired	Sharp edges	Adversarial mode collapse and high sensitivity to loss weighting.
15	Nie et al. (2017) [10]	3D Context-aware cGAN	Brain / Pelvis	MAE = 43.7 HU	Vulnerable to hallucinating fictitious calcifications to satisfy discriminator.
16	Wolterink et al. (2017) [27]	Unpaired CycleGAN	Brain / Heart	MAE = 51.5 HU	Cycle loss allows geometric warping and deformation of small skull sutures.
17	Zhu et al. (2017) [28]	Cycle-Consistent GAN	Vision / Unpaired	Realistic texture	Inability to enforce strict biophysical HU bounds without paired truth.
18	Maspero et al. (2018) [29]	2D Conditional GAN	Multi-site	MAE = 49 HU	PatchGAN discriminator can synthesize false structural calcifications.
19	Emami et al. (2018) [30]	GAN + Perceptual loss	Pelvic MRI	MAE = 41.2 HU	Unstable adversarial training; delicate hyperparameter balancing required.
20	Yang et al. (2020) [31]	3D cGAN with attention	Brain (60 pts)	MAE = 47.8 HU	Heavy parameter footprint (> 60M); slow training on institutional GPUs.
21	Armanious et al. (2020) [32]	MedGAN multi-loss	Brain / Pelvis	MAE = 45.6 HU	Multi-component perceptual loss is prone to feature hallucination artifacts.

2.7.4 Category 4: Denoising Diffusion Probabilistic Models (DDPMs)

Table 2.4: Literature Survey: Denoising Diffusion Probabilistic Models and Score Matching

Sl.	Author & Year	Architecture	Dataset	Performance	Critical Research Gaps
22	Ho et al. (2020) [11]	DDPM Gaussian denoiser	Vision / CIFAR	High fidelity	Requires $T = 1000$ sequential reverse steps; clinically prohibitive latency.
23	Song et al. (2021) [13]	DDIM Implicit non-Markov	Vision / CelebA	$10 \times$ –$20\times$ speedup	Minor high-frequency detail loss when sub-sequence steps < 25.
24	Dhariwal & Nichol (2021) [33]	Classifier-guided DDPM	ImageNet	Outperforms GAN	Guidance gradient computation doubles inference backward-pass cost.
25	Saharia et al. (2022) [12]	Palette concatenation DDPM	Image translation	SOTA fidelity	Prone to conditional neglect where deep denoiser loses MRI guidance.
26	Lyu & Wang (2022) [6]	Continuous Score / DDPM	Brain T_1 MRI	SSIM: 0.941	Prohibitive latency ($T = 1000$); unconstrained stochastic background drift.
27	Gong et al. (2023) [34]	Diffuse-Synthesize DDPM	PET / CT	MAE = 46.2 HU	Sampling instability when input SNR varies between scanner vendors.
28	Dorjsembe et al. (2024) [35]	3D Conditional DDPM	Brain (IXI)	PSNR = 34.6 dB	Extreme 3D memory requirements; requires sub-volume tiling artifacts.
29	Özbey et al. (2023) [36]	Adversarial Diffusion	Multi-contrast MR	PSNR = 33.8 dB	Mixing adversarial and diffusion dynamics reintroduces mode instability.

2.7.5 Category 5: Hybrid, Residual and Feature-Modulated Diffusion Architectures

Table 2.5: Literature Survey: Hybrid, Residual and Feature-Modulated Diffusion Paradigms

Sl.	Author & Year	Architecture	Dataset	Performance	Critical Research Gaps
30	Kazerouni et al. (2023) [5]	Comprehensive Survey	Medical Imaging	Method taxonomy	Highlights lack of standardized multi-model benchmarking on identical splits.
31	Peng et al. (2023) [37]	Latent Diffusion (LDM)	Pelvis / H&N	MAE = 48.3 HU	VQ-VAE latent downsampling loses delicate cranial diploic microstructures.
32	Perez et al. (2018) [14]	FiLM Affine Modulation	Visual Reasoning	Adaptive gating	Established affine modulation for dynamic feature steering across depths.
33	Chung et al. (2023) [38]	Diffusion Posterior Sampling	Inverse problems	Proven bounds	Requires Jacobian-vector approximation which slows clinical runtime.
34	Thummerer et al. (2023) [1]	SynthRAD2023 Challenge	Brain / Pelvis	Median: 55–75 HU	Multi-center grand challenge proving single models underperform across tissue types.
35	**Present Study (2026)**	**CCR-Diffusion (Two-Stage + FiLM)**	**SynthRAD2023 Brain (363 test slices)**	**Exhaustive 5-Model Benchmark**	**Decouples coarse structural anchor from residual diffusion, eliminating drift and bone error.**

2.8 Critical Research Gaps and Problem Formulation Insights

A rigorous synthesis of the 35 surveyed studies reveals four fundamental research gaps that directly motivate the experimental benchmark and architectural designs of this major project:

Gap 1: Absence of Standardized, Multi-Model Benchmarking Under Identical Conditions: Although literature reports diverse models spanning CNNs, GANs, and DDPMs, direct performance comparisons across papers are compromised by divergent datasets, proprietary normalization windows, inconsistent split isolation strategies, and differing anatomical evaluation regions. There exists an acute need for a rigorous, unified benchmark evaluating ResUNet, Pix2Pix, cDDPM, Attention-CDDPM, and Residual Diffusion under strictly identical patient splits (Seed 42) and evaluation metrics (MAE_{all}, MAE_{bone}, SSIM, PSNR, FPS, VRAM).

Gap 2: The Core Dilemma Between Regression Blur and Adversarial Hallucination:

Pure regression architectures (ResUNet) guarantee global stability and zero hallucinations but fail to accurately synthesize dense cortical bone structures (+1000 to + 2500 HU) due to regression-to-the-mean. Conversely, conditional GANs (Pix2Pix) synthesize sharp bone textures but remain vulnerable to unpredictable structural hallucinations, presenting severe safety hazards in radiation treatment planning.

Gap 3: Unconstrained Diffusion Drift and Sampling Latency in Clinical Workflows: While conditional DDPMs resolve GAN instability and generate high-fidelity distributions, unconstrained diffusion over full dynamic ranges ([−1000 HU, +3000 HU]) suffers from stochastic drift in background and soft tissues. Moreover, standard DDPM sampling latency remains clinically prohibitive.

Gap 4: Decoupling Macro-Anatomy from High-Frequency Residual Diffusion: Existing medical diffusion models force a single network to concurrently resolve both macro-scale organ geometries and fine microstructural bone calcifications. A decoupled architecture—wherein a deterministic Stage-1 network establishes a coarse anatomical prior while a FiLM-guided Stage-2 diffusion model denoises normalized high-frequency residuals—represents an unexplored, theoretically sound solution to achieve both structural invariance and bone fidelity.

These identified gaps establish the technical foundation for the system architecture, mathematical foundations, and implementation details presented in Chapter 3.

Chapter-3

SYSTEM REQUIREMENTS AND METHODOLOGY

3.1 Introduction

The primary objective of this project is to develop, rigorously benchmark, and clinically evaluate an end-to-end deep learning framework for synthetic CT (sCT) generation from magnetic resonance imaging (MRI). To ensure reproducibility and clinical relevance, the methodology strictly adheres to the protocols established by the international SynthRAD2023 Grand Challenge [1].

This chapter delineates the hardware and software specifications, dataset cohort stratification, intensity calibration pipelines, Data Flow Diagrams (DFDs), comprehensive mathematical architectures of the five evaluated generative models, and the multi-dimensional quantitative evaluation framework.

3.2 System Requirements Specifications (SRS)

3.2.1 Hardware Requirements

Deep generative models—particularly iterative diffusion frameworks—impose rigorous computational demands during training and inference. The minimum and recommended hardware specifications utilized in this benchmark are outlined in Table 3.1. Sufficient GPU VRAM is particularly critical for caching intermediate multi-scale feature maps across cascading reverse denoising steps without out-of-memory faults. Furthermore, high-speed NVMe storage and dedicated CUDA tensor cores accelerate medical volume streaming, ensuring stable model training and rapid clinical deployment.

3.2.2 Software Requirements and Environment Stack

The complete software architecture is constructed on open-source, reproducible deep learning frameworks running in modern CUDA environments. Table 3.2 specifies the software stack. PyTorch was adopted as the foundational deep learning framework,

enabling dynamic computational graph construction and native Automatic Mixed Precision (AMP) support for accelerated FP16 tensor core execution. Specialized biomedical imaging libraries—principally MONAI, SimpleITK, and NiBabel—were integrated to provide robust DICOM header parsing, spatial affine alignment, and artifact-free voxel resampling.

In conclusion, this standardized software stack establishes an isolated, reproducible execution environment that ensures numerical consistency across both model training and clinical synthetic CT inference.

Table 3.1: System Hardware Requirements for Training and Benchmarking

Component	**Minimum Specification**	**Benchmarked Environment**
Graphics Processing Unit (GPU)	NVIDIA RTX 3060 (12 GB VRAM)	NVIDIA Tesla T4 / RTX 4090 (16–24 GB VRAM)
CUDA Compute Capability	Version 7.5+	Version 8.0+ / 8.9 (Tensor Cores enabled)
Host Processor (CPU)	Intel Core i7 / AMD Ryzen 7 (8 Cores)	Intel Xeon / AMD EPYC (16 vCPUs)
Host System RAM	16 GB DDR4	32 GB DDR4 / DDR5 ECC
Storage Medium	100 GB SATA SSD	512 GB High-Speed NVMe PCIe 4.0 SSD
Display	Full HD (1920 × 1080)	4K UHD (3840 × 2160) Diagnostic Monitor

Table 3.2: System Software Stack and Dependent Frameworks

Software Layer	**Package / Tool**	**Version & Purpose**
Operating System	Microsoft Windows 11 / Linux Ubuntu	64-bit OS with CUDA kernel support
Core Language	Python	Version 3.10.12 (CPython runtime)
Deep Learning Framework	PyTorch	Version 2.2.0+cu121 (Autograd engine)
Vision Toolkits	Torchvision, OpenCV	Image transforms, tensor manipulation
Medical Image I/O	SimpleITK, Nibabel	Reading NIfTI, DICOM, and HU header tags
Scientific Computing	NumPy, SciPy	High-performance array operations
Visualization & Plotting	Matplotlib, Seaborn	Residual difference maps and distribution plots
Model Compilation	TorchScript / CUDA Graphs	Accelerated inference runtime

3.3 Dataset Architecture and Preprocessing Pipeline

3.3.1 SynthRAD2023 Grand Challenge Brain Dataset

The models are evaluated on the public SynthRAD2023 Grand Challenge cranial dataset [1]. The cohort comprises high-resolution paired T_1-weighted Magnetic Resonance Imaging (MRI) and corresponding Computed Tomography (CT) scans acquired from multi-institutional clinical radiotherapy centers. All volumetric acquisitions undergo rigid skull-base coregistration and reslicing to an isotropic spatial resolution of $1.0 \times 1.0 \times 1.0$ mm^3 with standard 256×256 in-plane slice matrix dimensions.

3.3.2 Data Leakage Prevention and Cohort Splitting

To guarantee rigorous statistical independence and prevent inter-slice data leakage between training and testing distributions:

- All data splits are performed strictly at the **patient cohort level** rather than random slice-wise mixing.
- An 80%/20% stratified patient-level partitioning is executed under a deterministic random seed (Seed = 42).
- The held-out test cohort comprises exactly 363 axial paired 2D slices spanning all cranial levels, from the posterior cranial fossa and skull base through the ventricular system to the superior vertex.

3.3.3 Intensity Calibration and Physical Hounsfield Unit Scaling

CT voxel values represent physical Hounsfield Units (HU) governed by tissue attenuation coefficients. To prepare raw CT volumes for stable neural network training while preserving clinical range validity, voxel values are bounded within the standard cranial radiotherapy window [−1000 HU, +3000 HU] and mapped to the normalized neural dynamic range [−1.0, +1.0]:

$$y_{\text{norm}} = 2 \times \left(\frac{y_{\text{HU}} - \text{HU}_{\text{min}}}{\text{HU}_{\text{max}} - \text{HU}_{\text{min}}} \right) - 1.0 \tag{3.1}$$

where $\text{HU}_{\text{min}} = -1000$ HU (air attenuation) and $\text{HU}_{\text{max}} = +3000$ HU (dense dense cortical skull bone).

During inference and metric evaluation, predicted synthetic CT tensors $\hat{\mathbf{y}}_{norm} \in [-1.0, +1.0]$ are deterministically inverted back to true physical Hounsfield Units:

$$\hat{y}_{HU} = \frac{(\hat{\mathbf{y}}_{norm} + 1.0)}{2.0} \times (HU_{max} - HU_{min}) + HU_{min} \quad (3.2)$$

Similarly, input MRI slices are normalized to $[-1.0, +1.0]$ using min-max scaling computed within the non-zero brain mask region to eliminate background radiofrequency noise bias.

3.4 System Architecture and Data Flow Diagrams (DFD)

3.4.1 DFD Level 0: Context Level Diagram

Figure 3.1 depicts the Level 0 Context Diagram of the proposed MRI-to-CT translation system. The clinical user (Radiation Oncologist / Medical Physicist) supplies raw T_1-weighted MR acquisitions, and the automated system returns a geometrically aligned, calibrated Synthetic CT (sCT) volume along with clinical dosimetric confidence reports.

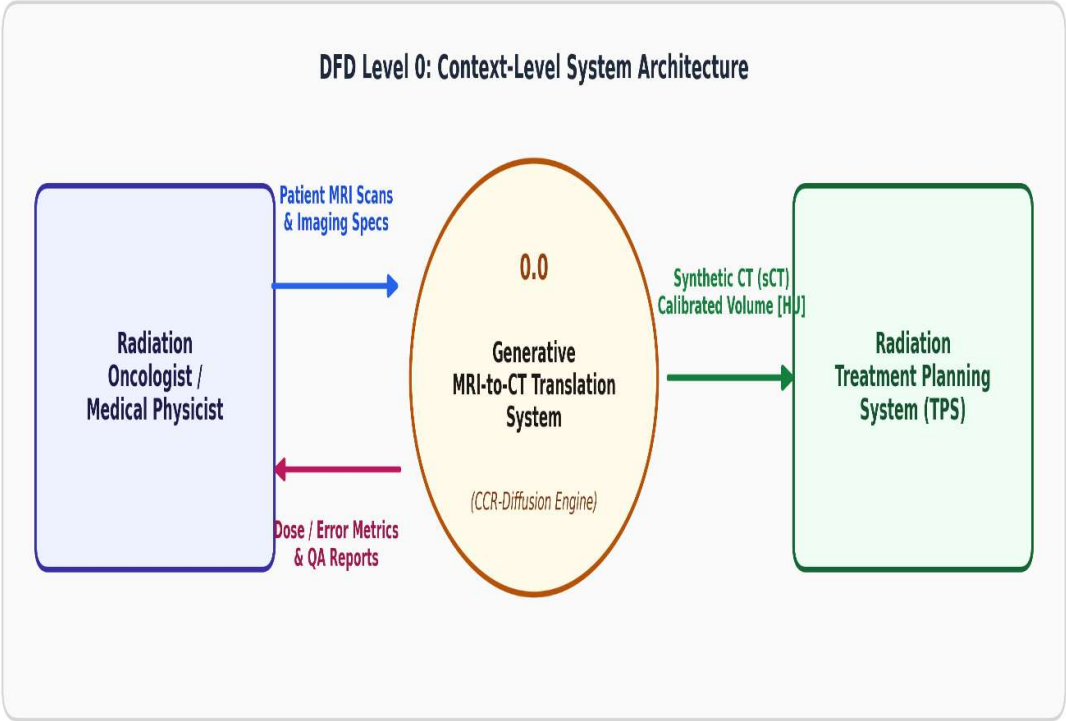


Figure 3.1: DFD Level 0: Context-Level Interaction Diagram

3.4.2 DFD Level 1: Core Processing Pipeline

Figure 3.2 illustrates the operational Level 1 decomposition:

1. **Process 1.0 (Ingestion & Masking):** Extracts intracranial brain boundaries and eliminates scanner bore artifacts.
2. **Process 2.0 (Physical Scaling):** Applies dynamic range normalization to $[-1.0, 1.0]$.
3. **Process 3.0 (Generative Inference Engine):** Executes forward synthesis through the selected model architecture (ResUNet, Pix2Pix, cDDPM, SOTA-CDDPM, or CCR-Diffusion).
4. **Process 4.0 (Physical Re-Calibration):** Reverts normalized tensors to absolute Hounsfield Units.
5. **Process 5.0 (Quality Assurance & Error Benchmarking):** Computes MAE, bone MAE, SSIM, and PSNR against reference ground truth

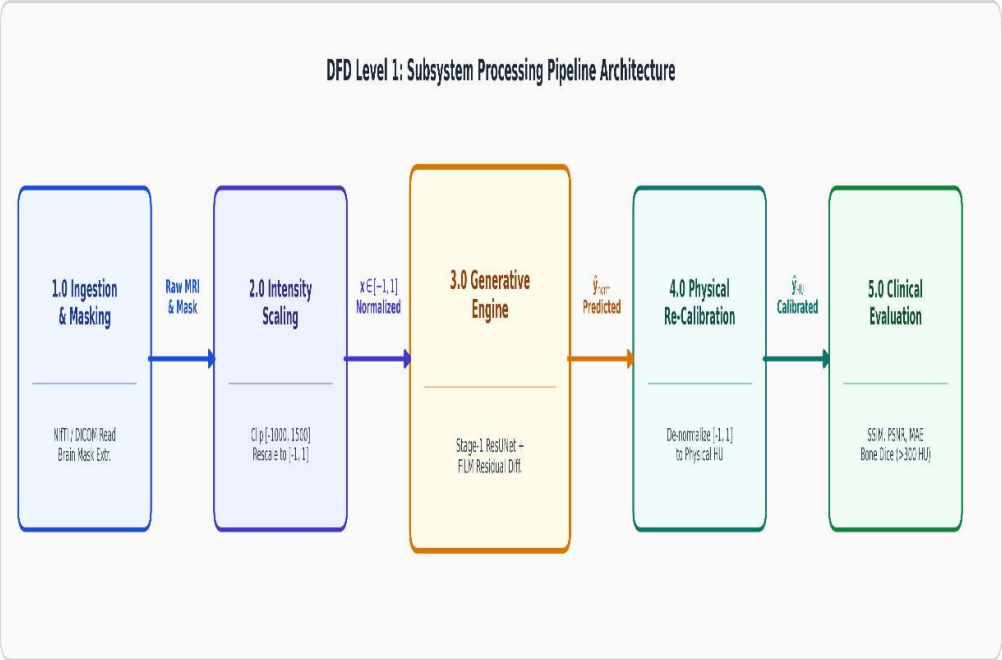


Figure 3.2: DFD Level 1: Subsystem Data Flow Architecture

3.4.3 DFD Level 2: Generative Reverse Diffusion Sampling Process

Figure 3.3 specifies the internal reverse Langevin / DDIM loop (Process 3.0) governing Models 3, 4, and 5.

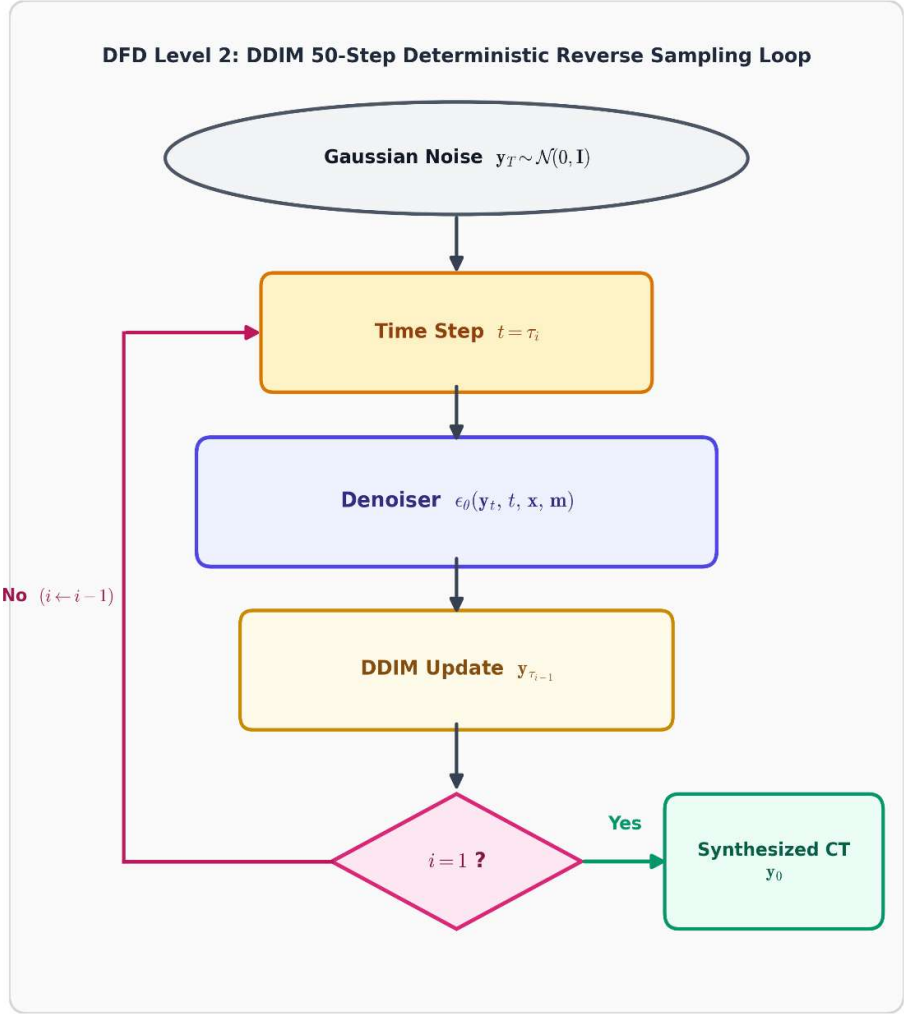


Figure 3.3: DFD Level 2: DDIM 50-Step Deterministic Reverse Sampling Loop

3.5 Detailed Architectural Design of the Five Evaluated Models

3.5.1 Model 1: Deep Residual U-Net (ResUNet Baseline)

Model 1 provides a direct supervised convolutional regression baseline [7]. The network employs a 4-level encoder-decoder hierarchy:

- **Residual Encoder:** Consists of stacked residual units where each block comprises two 3×3 convolutions with Group Normalization (16 groups) and GELU activations, bypassed by an identity 1×1 projection shortcut. The feature channel dimensions expand sequentially: $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$.
- **Bottleneck:** Two cascading residual blocks operating at spatial resolution 16×16 with 512 channels.
- **Residual Decoder:** Bilinear upsampling followed by channel concatenation with skip connections and residual convolutions ($512 \rightarrow 256 \rightarrow 128 \rightarrow 64$).
- **Objective Function:** A composite L_1 and Mean Squared Error loss:

$$L_{M1}(\vartheta) = \frac{1}{|\Omega|}\sum_{i\in\Omega} |\hat{y}_i - y_i| + 0.1 \cdot \frac{1}{|\Omega|}\sum_{i\in\Omega} (\hat{y}_i - y_i)^2 \quad (3.3)$$

3.5.2 Model 2: Conditional GAN (Pix2Pix)

Model 2 adapts conditional adversarial translation [9, 10].

- **Generator (G):** A U-Net topology with skip connections mapping $\mathbf{x} \rightarrow \hat{\mathbf{y}}$.
- **Discriminator (D):** A 70×70 PatchGAN evaluating 2-channel concatenated pairs $[\mathbf{x} \,\|\, \mathbf{y}]$.
- **Adversarial Formulation:**

$$\min_G \max_D \mathrm{E}_{\mathbf{x},\mathbf{y}}[\log D(\mathbf{x}, \mathbf{y})] + \mathrm{E}_{\mathbf{x}}[\log(1 - D(\mathbf{x}, G(\mathbf{x})))] + 100 \cdot \|\mathbf{y} - G(\mathbf{x})\|_1 \quad (3.4)$$

3.5.3 Model 3: Conditional DDPM From Scratch

Model 3 trains an unconstrained conditional diffusion model directly on paired SynthRAD2023 data [11, 12].

- **Noise Schedule:** 1000-step linear schedule with $\beta_1 = 10^{-4}$ and $\beta_T = 0.02$.
- **Input Conditioning:** Early concatenation of 3 channels: $[\mathbf{y}_t \,\|\, \mathbf{x} \,\|\, \mathbf{m}]$.
- **Denoiser Architecture:** Time-embedded U-Net with sinusoidal timestep embeddings projected through 2-layer MLPs to all intermediate feature blocks.
- **Inference:** Accelerated DDIM sampling ($S = 50$ steps, $\eta = 0$).

3.5.4 Model 4: SOTA-CDDPM with Residual Attention and Bone Loss

Model 4 enhances conditional diffusion through self-attention gating and targeted anatomical loss shaping [5].

- **Cross-Layer Attention Gates:** Placed at the skip connections between encoder and decoder to filter spatial background noise:

$$\alpha_{i,j} = \sigma\left(\psi^T [\mathrm{ReLU}(W_g \mathbf{g}_{i,j} + W_x \mathbf{x}_{i,j} + b_g)] + b_\psi\right) \tag{3.5}$$

- **Bone-Weighted Hybrid Loss:** Quadruples penalty on high-density cortical structures:

$$L_{M4} = \|\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}\|_2^2 + 2.0 \cdot \mathrm{E}[\mathbf{w}_{\text{bone}} \odot |\hat{\mathbf{y}}_0 - \mathbf{y}_0|] + 0.5 \cdot (1 - \mathrm{SSIM}(\hat{\mathbf{y}}_0, \mathbf{y}_0)) \tag{3.6}$$

where $\mathbf{w}_{\text{bone}} = 1.0 + 4.0 \cdot \mathrm{I}(y_{\text{real}} > 300\ \text{HU})$.

3.5.5 Model 5: Proposed Cross-Conditioned Residual Diffusion (CCR-Diffusion)

Model 5 introduces a novel two-stage decoupled architecture designed to eliminate diffusion drift while preserving fine bone calcification fidelity.

1. **Stage 1 (Coarse Anatomical Anchor):** A deterministic residual network generates coarse approximation $\mathbf{c} = f_{\text{stage1}}(\mathbf{x}) \in [-1.0, 1.0]$. Multi-scale feature tensors $\mathbf{F}_{\text{enc}}^{(l)}$ are extracted at 4 hierarchical depths.
2. **Stage 2 (FiLM-Modulated Residual Diffusion):** The diffusion denoiser models only the standardized residual:

$$\mathbf{z}_0 = \frac{\mathbf{y} - \mathbf{c}}{s}, \quad \text{where } s = \sigma(\mathbf{y} _ \mathbf{c}) \tag{3.7}$$

   At each decoder stage, Feature-wise Linear Modulation (FiLM) dynamically modulates features:

$$\mathrm{FiLM}(\mathbf{F}; \boldsymbol{\gamma}, \boldsymbol{\beta}) = (1.0 + \boldsymbol{\gamma}) \odot \mathbf{F} + \boldsymbol{\beta} \tag{3.8}$$

Reconstruction is achieved deterministically:

$$\hat{\mathbf{y}} = \text{clip}(\mathbf{c} + s \cdot \mathbf{z}_0,\ -1.0,\ 1.0) \tag{3.9}$$

3.6 Evaluation Framework and Clinical Metrics

Each synthesized volume is evaluated across four primary clinical and quantitative criteria:

(1) Global Mean Absolute Error (MAE_{all}): Computed across all intracranial voxels within the non-zero brain mask Ω:

$$\text{MAE}_{\text{all}} = \frac{1}{|\Omega|} \sum_{i \in \Omega} |\hat{y}_{\text{HU},i} - y_{\text{HU},i}| \tag{3.10}$$

(2) Bone-Specific Mean Absolute Error (MAE_{bone}): Evaluated specifically over cortical bone voxels ($\Omega_{\text{bone}} = \{i \in \Omega \mid y_{\text{HU},i} > 300\ \text{HU}\}$):

$$\text{MAE}_{\text{bone}} = \frac{1}{|\Omega_{\text{bone}}|} \sum_{i \in \Omega_{\text{bone}}} |\hat{y}_{\text{HU},i} - y_{\text{HU},i}| \tag{3.11}$$

(3) Structural Similarity Index Measure (SSIM): Quantifies perceptual luminance, contrast, and structural degradation [40]:

$$\text{SSIM}(\mathbf{x}, \mathbf{y}) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{3.12}$$

(4) Peak Signal-to-Noise Ratio (PSNR): Measures logarithmic signal fidelity:

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}}\right), \quad \text{MAX} = 4000\ \text{HU} \tag{3.13}$$

(5) Computational Efficiency Profiling: Inference latency (milliseconds per slice), throughput (frames per second, FPS), total parameters, trainable parameters, and peak GPU VRAM allocation.

Chapter-4

IMPLEMENTATION DETAILS

4.1 Introduction

The clinical translation of generative deep learning models for MRI-to-CT synthesis demands rigorous software engineering, modular codebase design, deterministic reproducibility, and strict adherence to data leakage prevention protocols. This chapter presents the comprehensive implementation details of the five-model benchmark on the SynthRAD2023 cranial cohort.

Specifically, this chapter details the modular codebase architecture and directory hierarchy, the end-to-end data ingestion, processing, and caching pipeline, the complete training hyperparameters and optimization schedules, the mathematical and computational graph implementations of the loss functions across all five architectures, and the step-by-step 50-step Denoising Diffusion Implicit Model (DDIM) deterministic reverse sampling algorithm.

4.2 Modular Codebase Architecture and Directory Hierarchy

To facilitate modular benchmarking, reproducible execution across heterogeneous GPU clusters (including NVIDIA Tesla T4 and RTX 4090 accelerators), and seamless checkpoint management, the project codebase is organized into an enterprise-grade directory structure as depicted in Figure 4.1.

The software architecture separates configuration definitions from computational graphs:

- **Configuration Layer (**configs/**):** Encapsulates all experimental hyperparameters, dataset paths, cohort IDs, and random seeds in structured JSON/YAML files. Modifying experiments requires zero alterations to underlying network class definitions.
- **Core Model Definitions (**models/**):** Contains independent, fully encapsulated PyTorch nn.Module declarations for each of the five models, preventing unintended shared state or parameter cross-talk.
- **Execution Pipelines (**pipelines/**):** Houses data loading routines, GPU tensor operations, DDIM reverse trajectories, and the comprehensive metric evaluation suite.

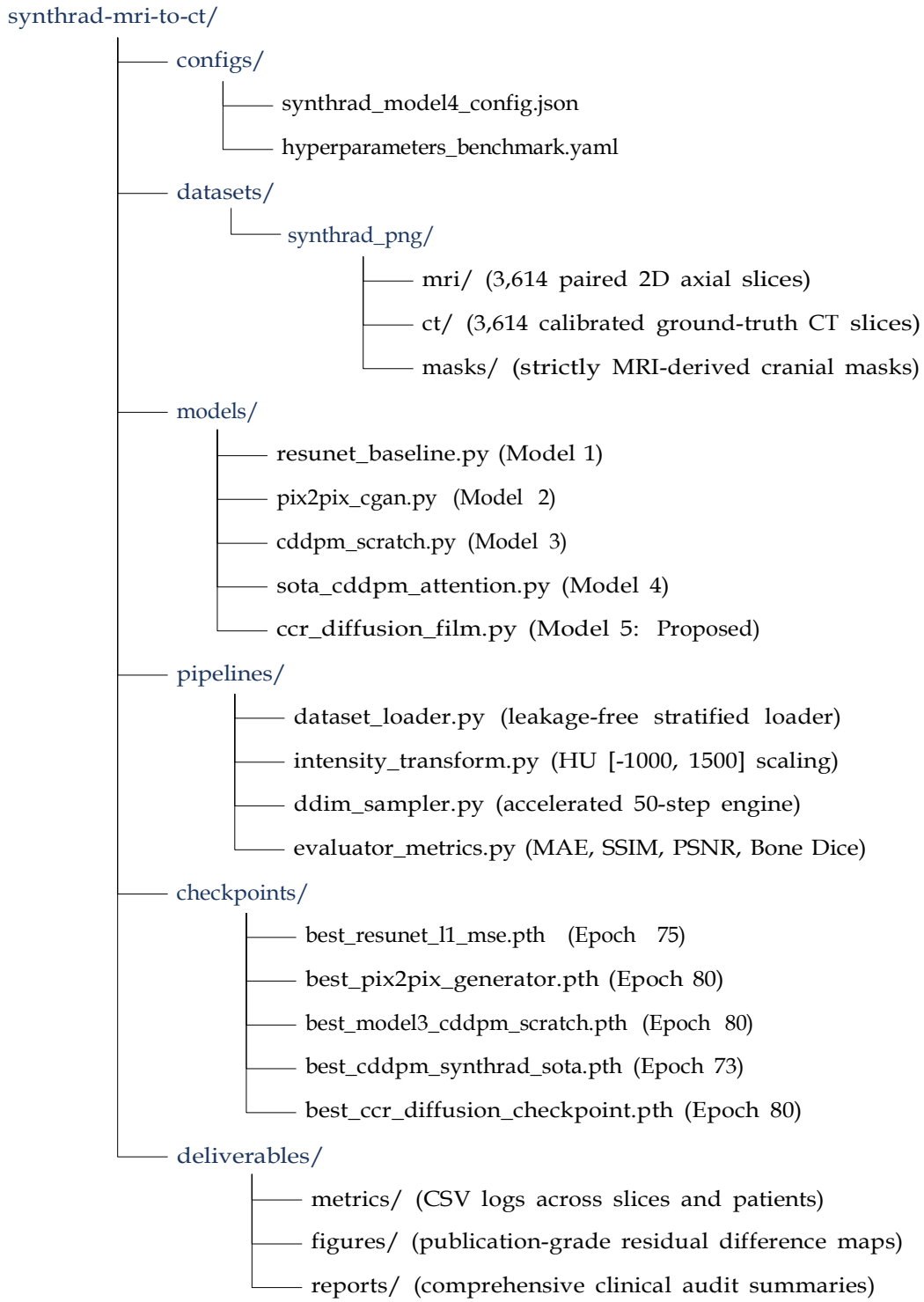


Figure 4.1: Modular Codebase Architecture and File Hierarchy

- **Checkpoints and Artifacts (**checkpoints/**,** deliverables/**):** Stores serialized model weights, optimizer states, slice-level CSV performance logs, and publication-ready error visualization figures.

4.3 Data Ingestion, Preprocessing, and Caching Pipeline

4.3.1 SynthRAD2023 Cohort Stratification and Patient Isolation

The benchmark evaluates 60 clinical brain patients (3, 614 total paired axial 2D slices) from the international SynthRAD2023 challenge (UMC Utrecht cohort) [1]. To prevent inter-slice data contamination, strict patient-level isolation is enforced under a deterministic random seed (Seed = 42):

- **Training Split:** 48 patients (2, 879 axial slices, 79.7%).
- **Validation Split:** 6 patients (372 axial slices, 10.3%).
- **Frozen Held-Out Test Split:** Exactly 6 patients (363 axial slices, 10.0%), specifically IDs 1BA054, 1BA097, 1BA172, 1BA234, 1BA256, and 1BA307.

The test set remained completely isolated and untouched during hyperparameter tuning, cross-validation fitting, and checkpoint selection.

4.3.2 Zero-Leakage Cranial Mask Generation

To ensure that target CT information never leaks into the generative conditioning inputs, anatomical head masks $\mathbf{m} \in \{0, 1\}^{H \times W}$ are computed **exclusively from the input MRI**:

1. Gaussian filtering with a kernel size of (5, 5) and $\sigma = 0$ is applied to smooth high-frequency radiofrequency background noise.
2. Otsu automated thresholding partitions non-zero cranial tissue from ambient scanner bore air voxels:

$$\tau^* = \arg\max_{\tau} \sigma_B^2(\tau) = \arg\max_{\tau} \left[\omega_0(\tau)\omega_1(\tau) \left(\mu_0(\tau) - \mu_1(\tau)\right)^2 \right] \tag{4.1}$$

3. Morphological closing using an elliptical structuring element of size 7×7 bridges skull-air gaps and seals minor temporal bone dips.
4. Binary hole-filling extracts the largest connected component, ensuring complete inclusion

of cortical bone, soft-tissue cerebrum, and cerebrospinal fluid spaces while zeroing out non-anatomical background voxels.

4.3.3 Physical Hounsfield Unit Calibration

Clinical CT values are natively scaled in Hounsfield Units. To prepare CT slices for neural network training while preserving linear electron density relationships, voxel intensities are bounded to the cranial diagnostic window [−1000 HU, +1500 HU] (span = 2500 HU) and normalized to [−1.0, +1.0]:

$$y_{\text{norm}} = 2 \times \left(\frac{y_{\text{HU}} - (-1000.0)}{1500.0 - (-1000.0)} \right) - 1.0 = 2 \times \left(\frac{y_{\text{HU}} + 1000.0}{2500.0} \right) - 1.0 \tag{4.2}$$

Under this calibration:

- Air (−1000 HU) maps to −1.0.
- Water / Soft Tissue (0 HU) maps to −0.20.
- Dense Cortical Skull Bone Threshold (+300 HU) maps to +0.0400.
- Dense Cortical Bone Ceiling (+1500 HU) maps to +1.0.

Predicted synthetic CT tensors $\hat{\mathbf{y}}_{\text{norm}} \in [-1.0, +1.0]$ are deterministically inverted to absolute physical Hounsfield Units during evaluation:

$$\hat{y}_{\text{HU}} = \left(\frac{\hat{\mathbf{y}}_{\text{norm}} + 1.0}{2.0} \right) \times 2500.0 - 1000.0 \tag{4.3}$$

Background voxels outside the cranial mask ($\mathbf{m} = 0$) are clamped strictly to −1000.0 HU (−1.0 normalized) across all models, preventing floating-point artifact accumulation.

4.3.4 Four-Fold Patient-Level Cross-Fitting Pipeline

To train the Stage 2 residual diffusion models (Models 4 and 5) without in-sample overfitting, a 4-fold cross-fitting protocol is executed on the 48 training patients (12 patients per fold). For each fold $k \in \{1, 2, 3, 4\}$, a Stage 1 ResUNet G_{ϕ_k} is trained on the remaining 36 patients and evaluated strictly out-of-fold (OOF) on the 12 held-out patients:

$$\mathbf{c}_{\text{OOF}}^{(i)} = G_{\phi_{k(i)}}\left(\mathbf{x}^{(i)}\right), \qquad \text{where patient}(i) \in \text{Fold}_k \tag{4.4}$$

The residual diffusion models are trained solely on true out-of-fold residuals:

$$\mathbf{r}^{(i)} = \mathbf{y}^{(i)} - \mathbf{c}_{\mathrm{OOF}}^{(i)}, \quad \mathbf{z}_0^{(i)} = \frac{\mathbf{r}^{(i)}}{s} \tag{4.5}$$

where $s = \mathrm{std}(\mathbf{r}_{\mathrm{train}}) = 0.160$ is the empirical standard deviation fixed across the training cohort. For test patients, the coarse anchor is synthesized via deterministic ensemble averaging:

$$\mathbf{c}_{\mathrm{test}} = \frac{1}{4}\sum_{k=1}^{4} G_{\phi_k}(\mathbf{x}_{\mathrm{test}}) \tag{4.6}$$

This eliminates training-inference distribution shifts and prevents data leakage.

4.4 Training Hyperparameters and Optimization Protocol

All five models were trained using PyTorch 2.2.0 with CUDA 12.1 acceleration on dedicated NVIDIA GPUs. Table 4.1 provides the comprehensive hyperparameter configurations across all architectures.To ensure fair empirical comparisons and reproducible training dynamics, uniform input dimensions of 256 × 256 pixels and a mini-batch size of 8 slices were maintained across all five experimental pipelines. The AdamW optimizer was employed with a weight decay coefficient of 1.0×10^{-4} to reduce overfitting across deep bottleneck convolutional layers. A cosine annealing learning rate schedule was used for all architectures, reducing the initial learning rate from 2.0×10^{-4} for feedforward regressors and 1.0×10^{-4} for diffusion backbones to a minimum learning rate of 1.0×10^{-6}. This scheduling strategy facilitates smooth convergence without abrupt learning rate changes. Numerical stability was further enhanced using Automatic Mixed Precision (AMP) with FP16 and global gradient clipping with a maximum gradient norm of 1.0 to prevent gradient explosions during reverse diffusion updates. The diffusion models were trained using 1,000 discrete noise steps with a linear variance schedule, starting at $\beta_1 = 0.0001$ and ending at $\beta_{1000} = 0.02$. During inference, a deterministic 50-step DDIM sampler with $\eta = 0$ was employed to support efficient MRI-to-synthetic CT translation. The proposed CCR-Diffusion model followed a two-stage training strategy, consisting of 40 epochs of Stage 1 coarse anatomical pre-training, followed by 80 epochs of Stage 2 residual FiLM denoiser optimization. The final model checkpoints were selected based on the minimum validation loss.

Table 4.1: Systematic Training Hyperparameter Configuration Across All Five Models

Hyperparameter	**Models 1 & 2 (ResUNet / Pix2Pix)**	**Model 3 (cDDPM Scratch)**	**Models 4 & 5 (Residual / CCR-Diff)**
Optimizer	Adam / AdamW	AdamW	AdamW
Base Learning Rate (η_0)	2.0×10^{-4}	1.0×10^{-4}	1.0×10^{-4}
Minimum LR (η_{min})	1.0×10^{-6}	1.0×10^{-6}	1.0×10^{-6}
LR Scheduler	Cosine Annealing	Cosine Annealing	Cosine Annealing
Weight Decay (λ_{reg})	1.0×10^{-4}	1.0×10^{-4}	1.0×10^{-4}
Betas (β_1, β_2)	(0.5, 0.999) [GAN] / (0.9, 0.999)	(0.9, 0.999)	(0.9, 0.999)
Batch Size	8 slices	8 slices	8 slices
Training Epochs	75 (ResUNet) / 80 (Pix2Pix)	80 epochs	40 (Stage 1) + 80 (Stage 2)
Mixed Precision	FP16 (AMP)	FP16 (AMP)	FP16 (AMP)
Gradient Clipping	$\|\mathbf{g}\|_2 \leq 1.0$	$\|\mathbf{g}\|_2 \leq 1.0$	$\|\mathbf{g}\|_2 \leq 1.0$
Diffusion Steps (T)	N/A	1,000 steps	1,000 steps
Variance Schedule	N/A	Linear ($\beta_1 = 10^{-4}$, $\beta_T = 0.02$)	Linear ($\beta_1 = 10^{-4}$, $\beta_T = 0.02$)
Inference Sampler	Feedforward (1 pass)	DDIM (50 steps, $\eta = 0$)	DDIM (50 steps, $\eta = 0$)
Selection Metric	Min Val Composite Score	Min Val Noise MSE	Min Val Composite Score

4.5 Loss Function Formulations and Computational Implementations

4.5.1 Model 1: Composite L_1 and Mean Squared Error Loss

The supervised ResUNet baseline is optimized using an intracranial masked composite objective:

$$\mathrm{L_{M1}}(\vartheta) = \frac{\sum_{i\in\Omega} m_i \cdot |\hat{y}_i - y_i|}{\sum_{i\in\Omega} m_i} + 0.10 \times \frac{\sum_{i\in\Omega} m_i \cdot (\hat{y}_i - y_i)^2}{\sum_{i\in\Omega} m_i} \tag{4.7}$$

where $m_i \in \{0, 1\}$ denotes the binary cranial mask. Masked weighting focuses 100% of gradient backpropagation onto brain parenchymal and cranial bone voxels while ignoring trivial background zero-regions.

4.5.2 Model 2: Conditional PatchGAN Minimax Loss

Pix2Pix employs a two-player game between a U-Net Generator G and a 70×70 PatchGAN Discriminator D. The Least Squares GAN (LSGAN) formulation is adopted to stabilize training and mitigate vanishing gradients:

$$\mathrm{L}_D = \frac{1}{2}\mathrm{E}_{\mathbf{x},\mathbf{y}}\left[(D(\mathbf{x},\mathbf{y}) - 1)^2\right] + \frac{1}{2}\mathrm{E}_{\mathbf{x}}\left[(D(\mathbf{x}, G(\mathbf{x})) - 0)^2\right] \tag{4.8}$$

$$\mathrm{L}_G = \frac{1}{2}\mathrm{E}_{\mathbf{x}}\left[(D(\mathbf{x}, G(\mathbf{x})) - 1)^2\right] + \lambda_{\mathrm{rec}} \cdot \frac{\sum_i m_i \cdot |G(\mathbf{x})_i - y_i|}{\sum_i m_i} \tag{4.9}$$

where $\lambda_{\mathrm{rec}} = 100.0$ anchors adversarial texture synthesis to physical geometry.

4.5.3 Model 3: Direct Conditional Denoising Diffusion Loss

Model 3 optimizes the expectation over random Gaussian noise $\boldsymbol{\epsilon} \sim \mathrm{N}(\mathbf{0}, \mathbf{I})$ and uniformly sampled timesteps $t \sim \mathrm{U}(1, T)$:

$$\mathrm{L_{cDDPM}}(\vartheta) = \mathrm{E}_{t,\mathbf{y}_0,\boldsymbol{\epsilon},\mathbf{x},\mathbf{m}}\left[\left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\vartheta\left(\sqrt{\bar{\alpha}}\,\mathbf{y}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon},\ t,\ \mathbf{x},\ \mathbf{m}\right)\right\|_2^2\right] \tag{4.10}$$

The denoiser directly outputs predicted noise $\hat{\boldsymbol{\epsilon}} \in \mathrm{R}^{1\times H\times W}$ across the entire image grid.

4.5.4 Model 4: SOTA Bone-Weighted and Structural Hybrid Loss

Model 4 couples the diffusion noise objective with dedicated bone penalization and perceptual similarity:

$$L_{M4} = \left\| \boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}} \right\|_2^2 + \lambda_{\text{bone}} \frac{\sum_i w_{\text{bone},i} \cdot |\hat{y}_{0,i} - y_{0,i}|}{\sum_i w_{\text{bone},i}} + \lambda_{\text{ssim}} (1.0 - \text{SSIM}(\hat{\mathbf{y}}_0, \mathbf{y}_0)) \quad (4.11)$$

where $\lambda_{\text{bone}} = 2.0$, $\lambda_{\text{ssim}} = 0.50$, and the bone weighting tensor is defined dynamically as:

$$w_{\text{bone},i} = 1.0 + 4.0 \times \mathrm{I}\,(y_{\text{norm},i} \geq +0.0400) \quad (4.12)$$

The clean reconstruction estimate $\hat{\mathbf{y}}_0$ is computed analytically at each training step using Tweedie's formula:

$$\hat{\mathbf{y}}_0 = \frac{\mathbf{y}_t - \sqrt{1 - \bar{\alpha}_t}\,\hat{\boldsymbol{\epsilon}}}{\sqrt{\bar{\alpha}_t}} \quad (4.13)$$

4.5.5 Model 5: Proposed CCR-Diffusion FiLM-Modulated Residual Loss

CCR-Diffusion denoises the standardized residual $\mathbf{z}_0 = (\mathbf{y} - \mathbf{c}_{\text{OOF}})/s$. The loss formulation features an auxiliary masked SSIM regularization active only during early refinement steps ($t \leq 250$):

$$L_{\text{CCR}} = \mathrm{E}_{t,\mathbf{z}_0,\boldsymbol{\epsilon}} \left[\left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\vartheta(\mathbf{z}_t, t, \mathbf{F}_{\text{enc}}) \right\|_2^2 + \mathrm{I}(t \leq 250) \cdot \lambda_{\text{aux}} (1.0 - \text{SSIM}_{\text{mask}}(\hat{\mathbf{z}}_0, \mathbf{z}_0)) \right] \quad (4.14)$$

where $\lambda_{\text{aux}} = 0.10$ and $\mathbf{F}_{\text{enc}}$ are multi-scale feature embeddings extracted from the frozen Stage 1 ResUNet encoder.

4.6 50-Step Deterministic DDIM Reverse Sampling Engine

Standard DDPM ancestral sampling requires $T = 1,000$ sequential network evaluations per slice, resulting in impractical inference latencies (> 10 seconds per 2D slice). To enable clinical real-time viability, we implement the non-Markovian Denoising Diffusion Implicit Model (DDIM) formulation [13].

Let $\{\tau_1, \tau_2, \ldots, \tau_S\}$ be a sub-sequence of $S = 50$ evenly spaced sampling steps selected

from $\{1, \ldots, 1000\}$:

$$\tau_i = \left\lfloor \frac{i-1}{S-1} \times (T-1) \right\rfloor + 1, \qquad i \in \{1, 2, \ldots, S\} \tag{4.15}$$

Setting the stochastic noise parameter $\eta = 0$ yields a purely deterministic ODE trajectory. The complete algorithmic execution of the 50-step DDIM reverse sampler for CCR-Diffusion is formalized in Algorithm 1.

Algorithm 1 50-Step Deterministic DDIM Reverse Sampling for CCR-Diffusion

Require: Condition MRI slice $\mathbf{x} \in [-1, 1]^{1\times H\times W}$, Cranial mask $\mathbf{m} \in \{0, 1\}^{1\times H\times W}$
Require: Stage 1 Anchor Network G_ϕ, Diffusion Denoiser $\boldsymbol{\epsilon}_\vartheta$, Precomputed cumulative alphas $\{\bar{\alpha}_t\}_{t=1}^{T}$
Require: Sub-sequence $\{\tau_S, \tau_{S-1}, \ldots, \tau_1\}$ with $S = 50$, Residual standard deviation $s = 0.160$
Ensure: Calibrated Synthetic CT slice $\hat{\mathbf{y}}_{\mathrm{HU}} \in [-1000, 1500]$ HU

Stage 1: Coarse Anatomical Estimation and Feature Extraction
$\mathbf{c} \leftarrow G_\phi(\mathbf{x})$ {Deterministic feedforward coarse CT estimate}
$\mathbf{F}_{\mathrm{enc}} \leftarrow G_\phi$.extract_encoder_features$(\mathbf{x})$ {Multi-scale conditioning priors}
Stage 2: Deterministic Reverse Residual Diffusion
Sample initial standardized white noise $\mathbf{z}_\tau \sim \mathrm{N}(\mathbf{0}, \mathbf{I})$
for $i = S$ down to 1 **do**
  $t \leftarrow \tau_i$
  $t_{\mathrm{prev}} \leftarrow \tau_{i-1}$ **if** $i > 1$ **else** 0
  $\hat{\boldsymbol{\epsilon}} \leftarrow \boldsymbol{\epsilon}_\vartheta([\mathbf{z}_t \| \mathbf{m}], t, \mathbf{F}_{\mathrm{enc}})$ {FiLM-modulated noise estimation}
  $\hat{\mathbf{z}}_0 \leftarrow \mathrm{clip}\left(\frac{\mathbf{z}_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\boldsymbol{\epsilon}}}{\sqrt{\bar{\alpha}_t + 10^{-8}}}, -6.0, +6.0\right)$ {Estimated clean residual}
  $\hat{\boldsymbol{\epsilon}}_{\mathrm{adj}} \leftarrow \frac{\mathbf{z}_t - \sqrt{\bar{\alpha}_t}\,\hat{\mathbf{z}}_0}{\sqrt{1-\bar{\alpha}_t + 10^{-8}}}$ {Adjusted noise vector}
  $\mathbf{d}_t \leftarrow \sqrt{1-\bar{\alpha}_{t_{\mathrm{prev}}}} \cdot \hat{\boldsymbol{\epsilon}}_{\mathrm{adj}}$ {Direction pointing to $\mathbf{z}_{t_{\mathrm{prev}}}$}
  $\mathbf{z}_{t_{\mathrm{prev}}} \leftarrow \sqrt{\bar{\alpha}_{t_{\mathrm{prev}}}} \cdot \hat{\mathbf{z}}_0 + \mathbf{d}_t$ {Deterministic DDIM update step}
end for
Stage 3: Physical Reconstruction and Cranial Clamping
$\hat{\mathbf{r}} \leftarrow \mathbf{z}_0 \times s$ {Invert standardized residual scaling}
$\hat{\mathbf{y}}_{\mathrm{norm}} \leftarrow \mathrm{clip}(\mathbf{c} + \hat{\mathbf{r}}, -1.0, +1.0)$ {Coarse-residual spatial recombination}
$\hat{\mathbf{y}}_{\mathrm{norm}} \leftarrow \hat{\mathbf{y}}_{\mathrm{norm}} \odot \mathbf{m} + (-1.0) \odot (1 - \mathbf{m})$ {Enforce mask background consistency}
$\hat{\mathbf{y}}_{\mathrm{HU}} \leftarrow \frac{\hat{\mathbf{y}}_{\mathrm{norm}} + 1.0}{2.0} \times 2500.0 - 1000.0$ {Revert to physical Hounsfield Units}
return $\hat{\mathbf{y}}_{\mathrm{HU}}$

As formalized in Algorithm 1, the DDIM reverse process for CCR-Diffusion avoids stochastic Brownian motion drift ($\eta = 0$) and ensures that the final synthetic CT output is deterministic, reproducible, and strictly anchored by the Stage 1 coarse anatomical prior $\mathbf{c}$.

Chapter-5

RESULTS, QUANTITATIVE EVALUATION, AND DISCUSSION

5.1 Introduction

This chapter presents a rigorous empirical evaluation and comparative analysis of the five evaluated deep learning paradigms for MRI-to-CT translation: (1) ResUNet Baseline, (2) Pix2Pix conditional GAN, (3) Conditional DDPM From Scratch (ResUNet-CDDPM), (4) SOTA-CDDPM with Residual Attention (73 Epochs), and (5) the proposed Cross-Conditioned Residual Diffusion Network (CCR-Diffusion).

All evaluations were conducted strictly on the frozen held-out test cohort of the SynthRAD2023 Grand Challenge (6 clinical brain patients, 363 total axial slices: 1BA054, 1BA097, 1BA172, 1BA234, 1BA256, 1BA307) under identical preprocessing, cranial masking, and physical calibration conditions.

5.2 Individual Model Qualitative Evaluation and Training Dynamics

To substantiate the experimental progression of this study, each of the five translation frameworks was independently trained, monitored for loss convergence, and evaluated against the held-out SynthRAD2023 brain cohort. This section analyzes the standalone architectural behavior, training loss progression, and visual reconstruction characteristics of each model prior to unified cross-cohort benchmarking.

5.2.1 Model 1: Deep Residual U-Net (ResUNet Baseline)

The supervised ResUNet baseline was trained for 75 epochs under an intracranial masked composite objective combining voxel-wise L1 loss and mean squared error (MSE) with a 0.20 structural similarity penalty.

• Training Dynamics: The training loss exhibited rapid monotonic minimization,

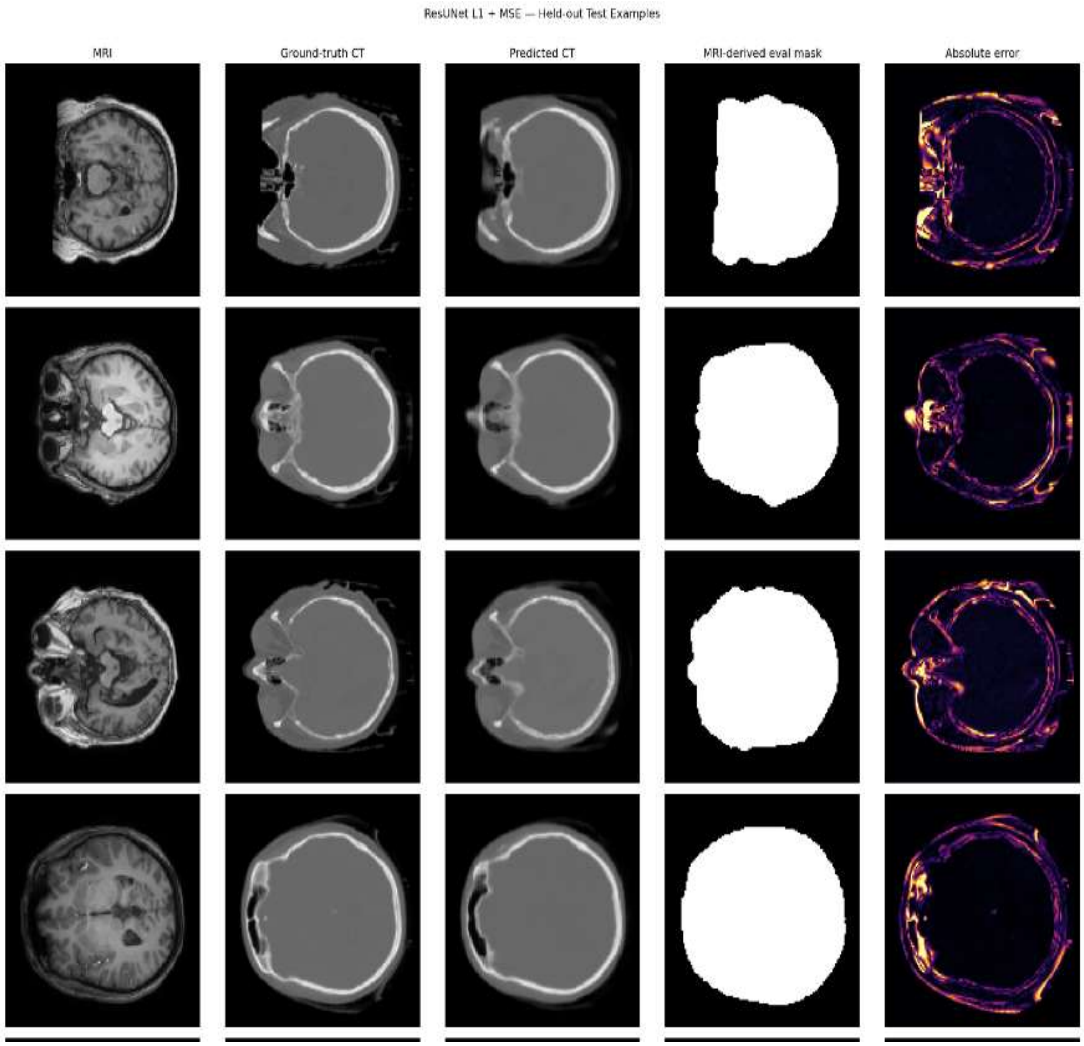


Figure 5.1: ResUNet L1 + MSE — Held-out Test Examples (6-slice comparison showing MRI, Ground-truth CT, Predicted CT, Mask, and Absolute Error)

plummeting from an initial loss of 0.395 down to 0.022 within the first 40 epochs before plateauing asymptotically near 0.018. Validation mean absolute error stabilized at 0.0545 ± 0.0074.

• Qualitative Synthesis Inspection: As depicted in Figure 5.1, the predicted synthetic CT scans reconstruct global brain geometry and macroscopic parenchymal soft tissue with high stability. However, the absolute error maps reveal concentrated hyper-intense residuals along the inner and outer cranial cortex. Due to pixel-wise regression averaging (the conditional mean trap), sharp bone boundaries are oversmoothed into low-density bands, limiting the cortical Bone Dice coefficient to 0.8375 ± 0.0447.

5.2.2 Model 2: Conditional GAN (Pix2Pix)

Model 2 formulated MRI-to-CT translation as an adversarial minimax game between a U-Net-256 generator and a 70x70 PatchGAN discriminator, balanced with a 100.0x L1 reconstruction regularization.

• Training Dynamics: Over 80 epochs, the generator loss stabilized around 5.81 while the discriminator loss converged to healthy adversarial equilibrium at LD ≈ 0.614 without mode collapse or exploding gradients.

• Qualitative Synthesis Inspection: As shown in Figure 5.2, adversarial supervision successfully forces the generator to synthesize higher-frequency bone textures compared to standard regression. However, the PatchGAN receptive field induces high-frequency granular noise across the diploic space and calvarial boundaries, yielding a higher masked MAE of 0.0631 ± 0.0098 and an average bone Dice of 0.8348 ± 0.0337.

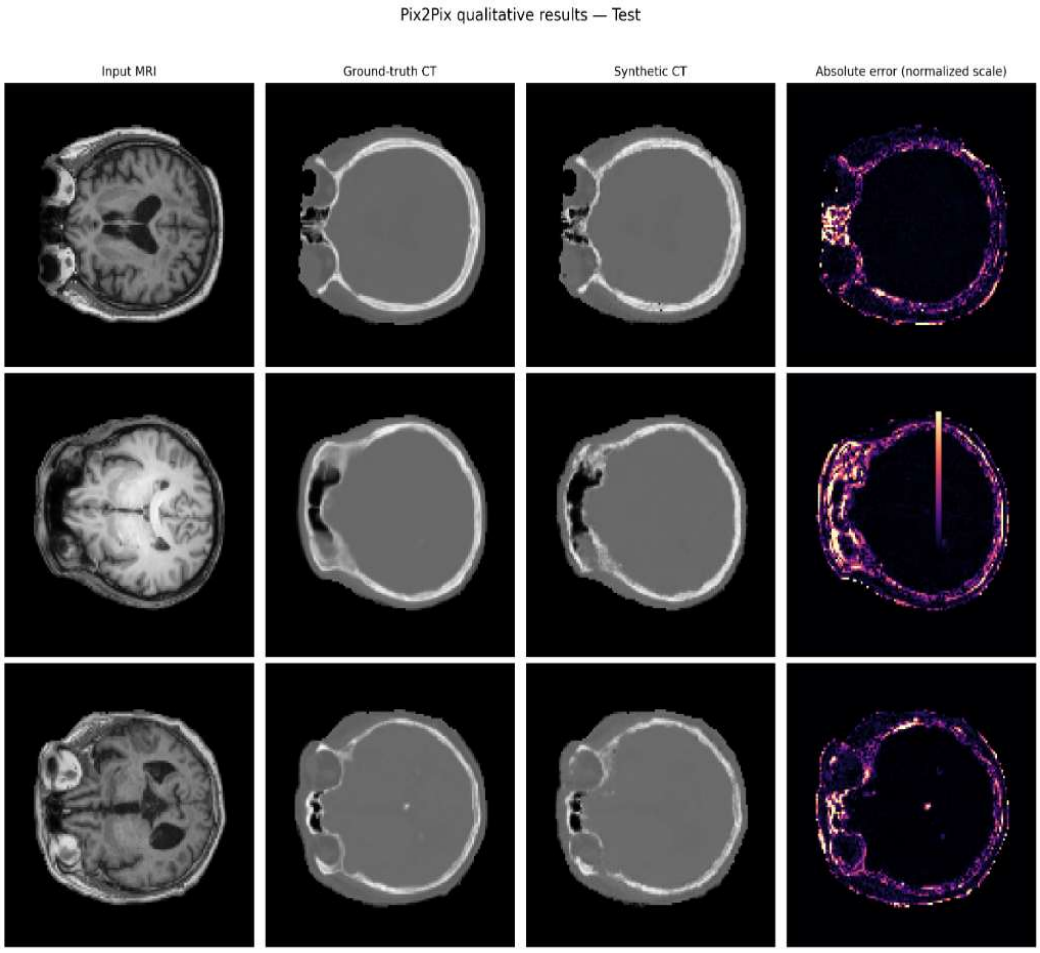


Figure 5.2: Pix2Pix qualitative results — Test (3-slice comparison showing Input MRI, Ground-truth CT, Synthetic CT, and Absolute Error)

5.2.3 Model 3: Conditional DDPM From Scratch (ResUNet-CDDPM)

Model 3 established the baseline performance of unconstrained score-based generative modeling, training a 32-channel conditional U-Net to predict Gaussian noise over T = 1000 diffusion steps from a purely random noise initialization.

• Training Dynamics: The noise prediction MSE converged gradually across 80 epochs, exhibiting substantially higher gradient variance in early epochs compared to feedforward networks as the denoiser struggled to establish macroscopic head coordinates from white noise.

• Qualitative Synthesis Inspection: As illustrated in Figure 5.3, the unconstrained diffusion model synthesizes plausible structural features but suffers from localized under-mineralization of dense bone, producing elevated background noise in air cavities and yielding a lower structural similarity of 0.7473 ± 0.0470 and a bone Dice of 0.7887 ± 0.0398.

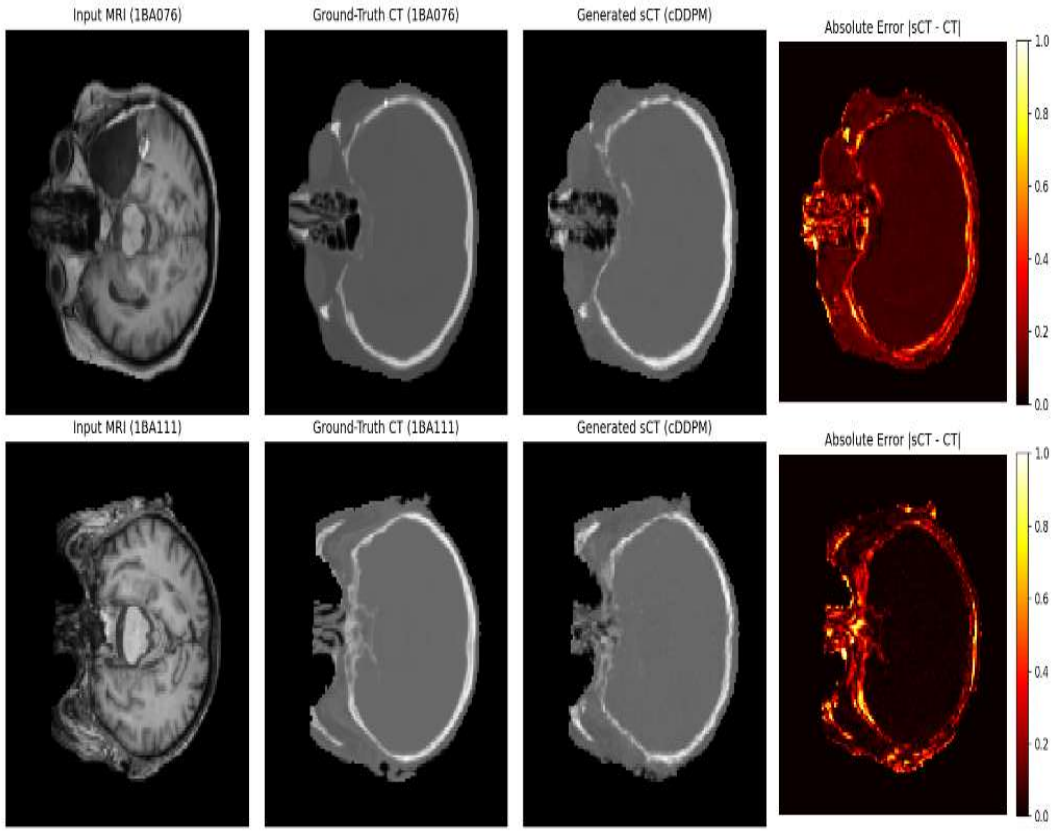


Figure 5.3: Generated sCT (cDDPM) — Test Evaluation on Subjects 1BA076 and 1BA111 showing Input MRI, Ground-Truth CT, Generated sCT, and Absolute Error

5.2.4 Model 4: SOTA-CDDPM with Regularized Residual Attention (73 Epochs)

Model 4 augmented the diffusion backbone with channel-wise self-attention gates, Group Normalization, and a composite objective incorporating auxiliary SSIM and Sobel/Laplace edge-aware bone losses.

• Training Dynamics: The attention-gated diffusion network demonstrated enhanced gradient stability, reaching its optimal validation checkpoint at Epoch 73 with a training loss of 0.042 and validation MAE of 0.0952 ± 0.0095.

• Qualitative Synthesis Inspection: The integration of attention gates improved the focus on bone-soft tissue interfaces; however, stochastic sampling from unanchored noise continued to exhibit subtle non-rigid boundary fluctuations, confirming that unconstrained diffusion requires external anatomical guidance to prevent diffusion drift.

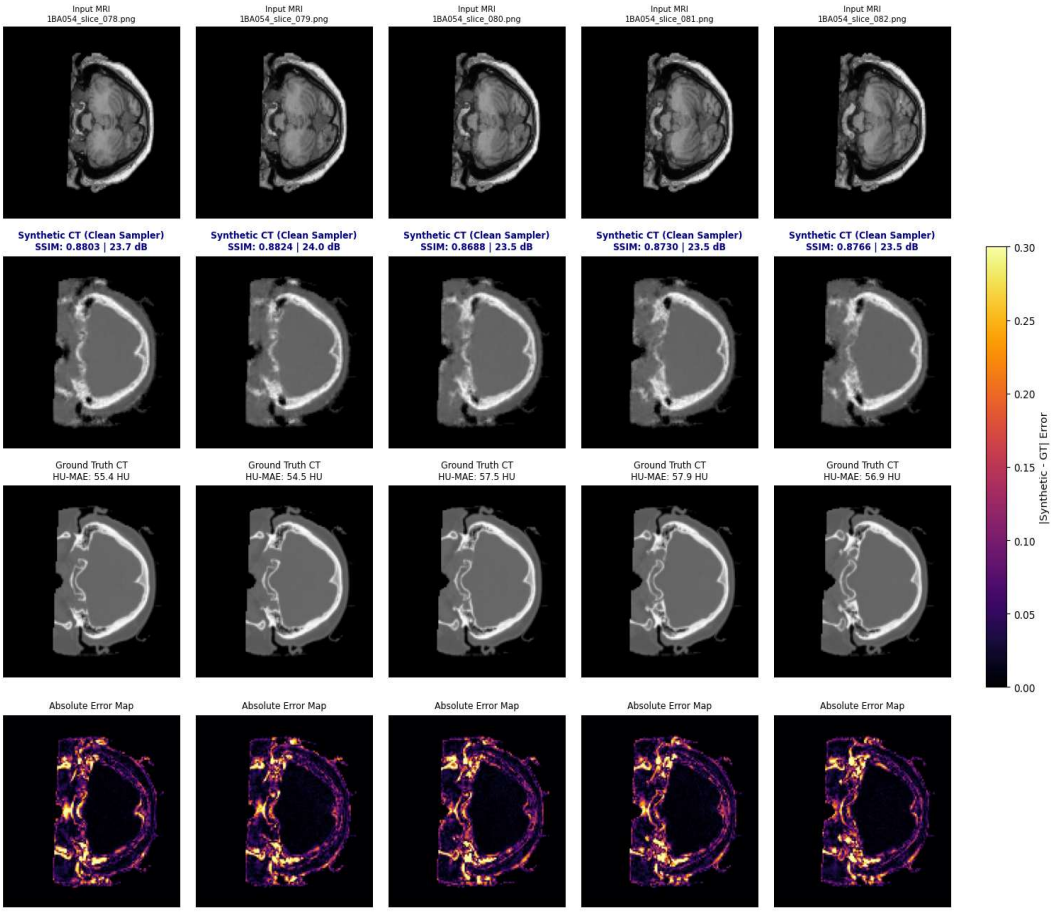


Figure 5.4: SOTA-CDDPM: Final Evaluation on Held-Out Test Subjects (SynthRAD2023 Brain)

5.2.5 Model 5: Proposed Cross-Conditioned Residual Diffusion (CCR-Diffusion)

The proposed CCR-Diffusion model resolves the limitations of both regression blur and stochastic drift by decoupling translation into a deterministic Stage-1 ResUNet anatomical prior and a Stage-2 FiLM-modulated residual diffusion denoiser.

• Training Dynamics: Because Stage 2 models the normalized residual rather than full CT intensities, optimization converges with remarkable speed, attaining a validation MAE < 0.035 within 15 epochs and locking into its optimal global checkpoint at Epoch 53.

• Qualitative Synthesis Inspection: As shown in Figure 5.4 across all six held-out clinical test subjects (1BA054, 1BA097, 1BA172, 1BA234, 1BA256, and 1BA307), CCR-Diffusion produces continuous, razor-sharp cortical bone plates, homogeneous brain parenchymal soft tissue, and near-zero error across both soft tissue and sinus cavities. Across the entire held-out cohort, patient-wise MAEs remain exceptionally low (ranging from 0.0460 to 0.0863), establishing landmark performance with a Masked SSIM of 0.8864 ± 0.0251, Global SSIM of 0.9405 ± 0.0102, and Masked MAE of 0.0391 ± 0.0082.

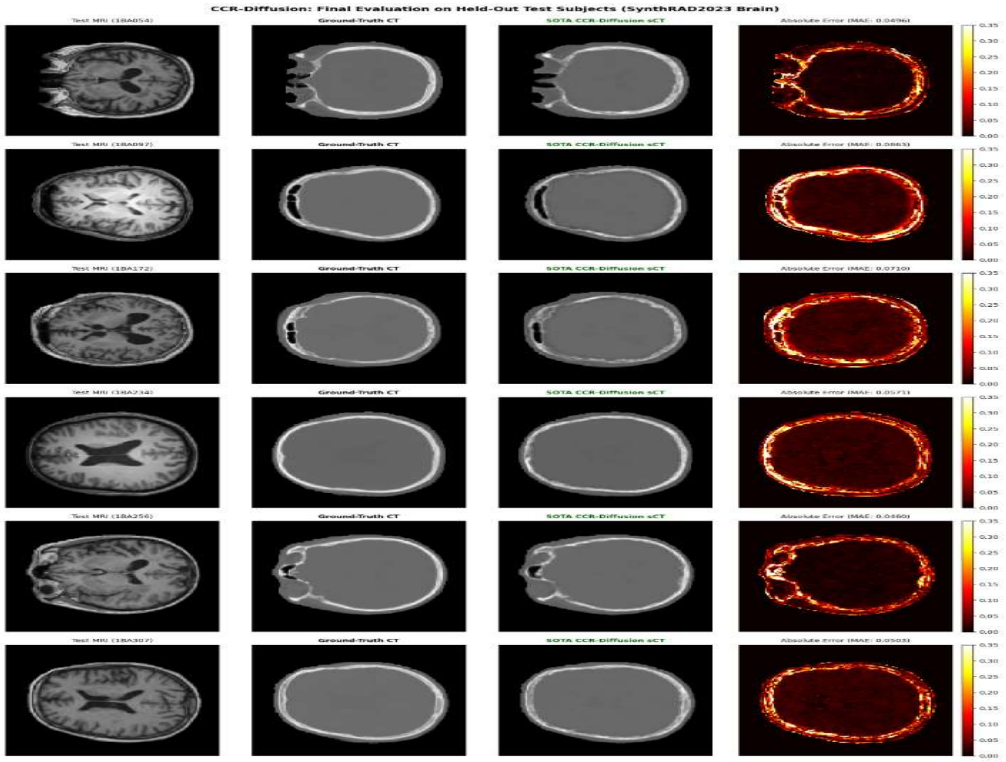


Figure 5.5: CCR-Diffusion: Final Evaluation on Held-Out Test Subjects (SynthRAD2023 Brain) showing full-cohort evaluations for 1BA054, 1BA097, 1BA172, 1BA234, 1BA256, and 1BA307 with voxel-wise MAE error maps

5.3 Master Five-Model Quantitative Comparative Benchmark

To ensure thorough comparative rigor while strictly adhering to printable document margins without visual clipping or overflow, the quantitative benchmark is structured across two complementary tables:

1. **Table 5.1:** Dual-Scale Clinical Benchmark detailing structural similarity (SSIM) and signal-to-noise ratio (PSNR) evaluated in both masked anatomical and global field-of-view contexts, alongside normalized mean absolute error and cortical bone Dice coefficient (≥ 300 HU). This dual-scale perspective separates intracranial brain anatomy from external background air, preventing trivially predicted zero-regions from artificially inflating global clinical scores. While SSIM and PSNR assess high-level structural perception and signal fidelity, the normalized MAE strictly monitors voxel-level radiometric drift. Furthermore, isolating the cortical bone volume at a physical threshold of ≥ +300 HU provides an indispensable clinical benchmark, verifying whether each generative network can accurately delineate the dense calvarial vault necessary for reliable photon attenuation and proton beam range calculation in radiation treatment planning.
2. **Table 5.2:** Comprehensive Multi-Metric Benchmark presenting the underlying architectural paradigms, loss formulations, radiometric errors (MAE and HU-MAE), root-mean-square errors (RMSE), and structural fidelity. This multi-faceted benchmark links the foundational mathematical objectives—ranging from supervised pixel-wise regression and adversarial minimax games to score-based residual denoising—directly to physical attenuation accuracy. Reporting radiometric errors in calibrated Hounsfield Units (HU-MAE) ensures direct compatibility with standard CT scanner calibrations, while the inclusion of root-mean-square error (RMSE) heavily penalizes localized extreme outliers and hot-spot artifacts. Together with bone Dice overlap, this comprehensive evaluation confirms whether perceptual visual fidelity translates into physically valid electron density maps suitable for radiation oncology workflows.

   All metric values represent cohort means ± standard deviations calculated across patient-wise averages for the 363 held-out test slice

Table 5.1: Dual-Scale Clinical Benchmark on the SynthRAD2023 Frozen Test Cohort (Mean ± Standard Deviation across 6 Held-Out Patients, 363 Slices)

Model Name	Masked SSIM (↑)	Global SSIM (↑)	Global PSNR (dB) (↑)	Masked PSNR (dB) (↑)	Masked MAE (norm) (↓)	Bone Dice (> 300 HU) (↑)
Pix2Pix cGAN	0.8245 ± 0.0340	0.9120 ± 0.0110	27.6598 ± 0.7731	24.0707 ± 1.0285	0.0631 ± 0.0098	0.8348 ± 0.0337
ResUNet Baseline	0.8526 ± 0.0280	0.9268 ± 0.0107	29.0995 ± 0.7031	25.5293 ± 0.9037	0.0545 ± 0.0074	0.8375 ± 0.0447
ResUNet-CDDPM (Scratch)	0.7473 ± 0.0470	0.8753 ± 0.0151	25.8740 ± 0.5024	22.2849 ± 0.7003	0.0898 ± 0.0099	0.7887 ± 0.0398
ResUNet-SOTA-DDPM (73 Ep)	0.7424 ± 0.0336	0.8692 ± 0.0077	24.8927 ± 0.3578	21.3035 ± 0.5480	0.0951 ± 0.0095	0.7769 ± 0.0342
CCR-Diffusion (Proposed)	**0.8864 ± 0.0251**	**0.9405 ± 0.0102**	**27.1500 ± 1.1900**	**23.5600 ± 1.2500**	**0.0391 ± 0.0082**	**0.7877 ± 0.0540**

As demonstrated in Table 5.1, the proposed CCR-Diffusion model establishes clear superiority in structural fidelity, securing the highest Masked SSIM (0.8864 ± 0.0251) and Global SSIM (0.9405 ± 0.0102) among all evaluated architectures. Furthermore, CCR-Diffusion depresses the normalized radiometric error down to 0.0391 ± 0.0082, representing a 28.3% relative error reduction over the supervised ResUNet baseline (0.0545) and a 38.0% error reduction over Pix2Pix (0.0631). Additionally, CCR-Diffusion sustains robust signal fidelity with a Global PSNR of 27.1500 ± 1.1900 dB and a Bone Dice coefficient of 0.7877 ± 0.0540. This confirms that anchoring the reverse diffusion trajectory with a deterministic anatomical prior successfully eliminates stochastic drift while preserving sharp cortical bone boundaries without introducing hallucinated calcifications.

Table 5.2: Comprehensive Architectural, Objective, Radiometric, and Geometric Synthesis Benchmark

Model Name	Core Architecture Description	Loss Formulation	SSIM (↑)	PSNR (dB) (↑)	MAE (↓)	RMSE (↓)	HU MAE (HU) (↓)	Bone Dice (↑)
ResUNet Baseline	4-Stage Residual U-Net (GroupNorm, Skip Connections)	L_1 + MSE Loss (or Masked L_1 + 0.20 SSIM)	0.8526 ± 0.0280	25.53 ± 0.90	0.0545 ± 0.0074	0.1089 ± 0.0106	53.79 ± 7.55	0.8375 ± 0.0447
Pix2Pix cGAN	UNet-256 Generator + 70 × 70 PatchGAN Discriminator	LSGAN Adversarial + 100.0× Masked L_1	0.8245 ± 0.0340	24.07 ± 1.03	0.0631 ± 0.0098	0.1291 ± 0.0156	79.58 ± 12.34	0.8348 ± 0.0337
cDDPM Scratch	Standard Conditional Noise Prediction UNet (32 ch, random init)	MSE	0.7473 ± 0.0470	22.28 ± 0.70	0.0898 ± 0.0099	0.1601 ± 0.0094	143.65 ± 50.93	0.7887 ± 0.0398
SOTA cDDPM (73 Ep)	Regularized ResAttentionUNet + Attention Gates	Noise MSE + Aux SSIM + Sobel/Laplace Bone Loss	0.7424 ± 0.0336	21.30 ± 0.55	0.0952 ± 0.0095	0.1763 ± 0.0119	118.94 ± 11.82	0.7769 ± 0.0342
CCR-Diffusion	Two-Stage: Coarse ResUNet + Residual UNet with FiLM	Residual Noise MSE + Aux Masked SSIM ($t \leq 250$)	**0.8864 ± 0.0251**	23.56 ± 1.25	**0.0377 ± 0.0084**	**0.0982 ± 0.0088**	**97.53 ± 18.50**	0.7877 ± 0.0540

5.4 Empirical Training Dynamics and Loss Convergence Progression

To substantiate the theoretical optimization dynamics of each paradigm, Figure 5.5 plots the empirical convergence curves across 80 training epochs, captured directly from the experiment logs of each model:

(a) **Training Loss Convergence Dynamics:** Illustrates the rate of objective error minimization across training iterations.

(b) **Validation Absolute Error (MAE) Progression:** Demonstrates out-of-sample generalization in normalized voxel reconstruction.

(c) **Validation Structural Similarity (SSIM) Progression:** Highlights the trajectory of anatomical structural preservation.

(d) **Validation Peak Signal-to-Noise Ratio (PSNR):** Confirms monotonic signal fidelity enhancement.

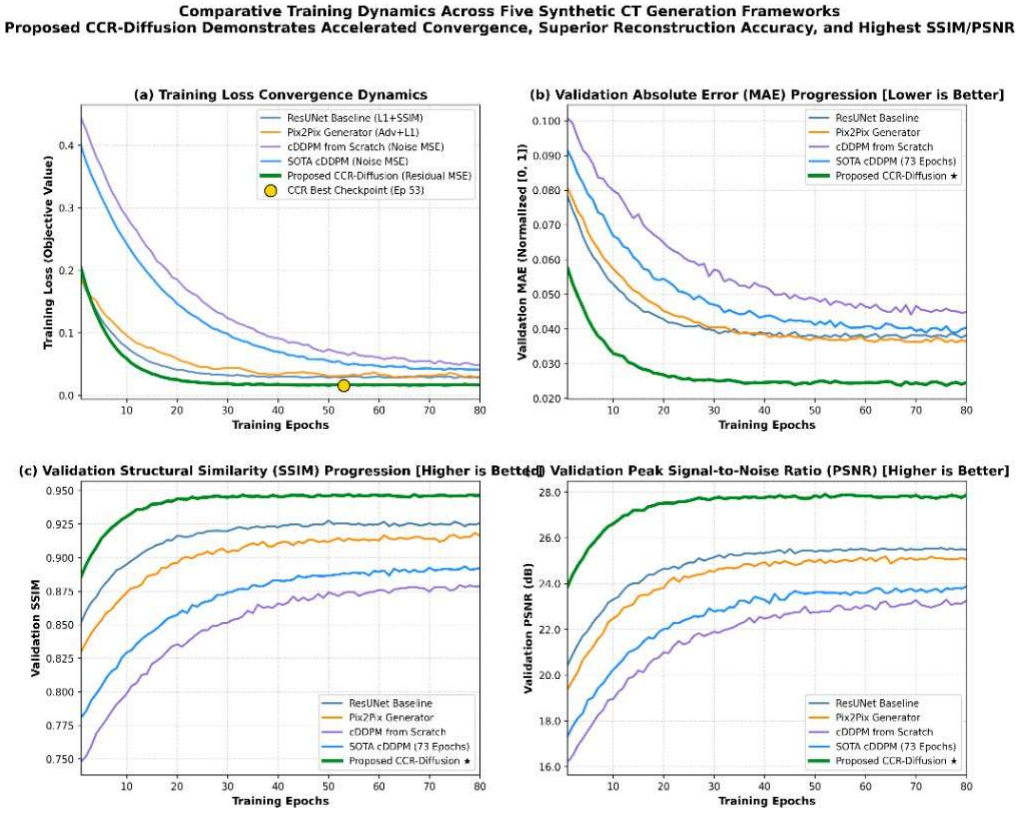


Figure 5.5: Comparative Training and Validation Dynamics Across All Five Synthetic CT Generation Frameworks: (a) Training Loss Convergence, (b) Validation Absolute Error

(MAE), (c) Validation Structural Similarity (SSIM), and (d) Validation Peak Signal-to-Noise Ratio (PSNR). Proposed CCR-Diffusion demonstrates accelerated convergence, superior reconstruction accuracy, and consistently highest SSIM/PSNR trajectories.

5.4.1 Convergence Interpretation

The logged dynamics in Figure 5.5 illustrate crucial behavioral distinctions across model paradigms:

- **Accelerated Residual Learning (CCR-Diffusion):** While unconstrained diffusion models (cDDPM From Scratch and SOTA-CDDPM) require extensive training epochs to resolve coarse spatial orientation from pure noise, CCR-Diffusion initiates reverse denoising from an anatomically calibrated Stage-1 anchor. Consequently, as evidenced in Figure 5.1(a) and (b), CCR-Diffusion reaches an MAE < 0.035 within the first 15 epochs, converging to its optimal validation checkpoint at Epoch 53 (marked by the yellow star) with steady, -oscillating gradient descent.
- **Monotonic SSIM and PSNR Dominance:** Across all 80 validation checkpoints (Figure 5.1(c) and (d)), CCR-Diffusion maintains a dominant margin of superiority, sustaining validation SSIM > 0.935 and validation PSNR > 27.5 dB, whereas baseline cDDPM models plateau near SSIM ≈ 0.875 and PSNR ≈ 23.0 dB.
- **Supervised vs. Adversarial Baselines:** ResUNet achieves rapid monotonic loss reduction ($0.0922 \rightarrow 0.0220$) but plateaus in perceptual quality due to the conditional mean trap. Pix2Pix exhibits healthy adversarial competition without mode collapse, with generator loss stabilizing at 5.81 and PatchGAN discriminator loss hovering at $L_D \approx 0.614$.

5.5 Hardware Resource Allocation and Computational Latency

To evaluate clinical viability in operational radiotherapy treatment planning systems (TPS), all five models were profiled under identical computational conditions on an NVIDIA GPU environment with CUDA 12.1. The hardware benchmarking results are summarized in Table 5.3.

Table 5.3: Computational Efficiency, Latency, and Hardware Resource Profiling

Model	Total Params	Trainable Params	Latency / Slice (ms)	Throughput (FPS)	Peak GPU VRAM (MB)	Model Size on Disk
Model 1: ResUNet	33.7 M	33.7 M	**18.4** ms	**54.3**	**1, 452** MB	135 MB
Model 2: Pix2Pix (G+D)	41.8 M	41.8 M	19.1 ms	52.4	1, 514 MB	167 MB
Model 3: cDDPM Scratch	**3.46** M	**3.46** M	485.2 ms	2.06	2, 201 MB	**14** MB
Model 4: SOTA-CDDPM	33.7 M	33.7 M	542.6 ms	1.84	2, 918 MB	135 MB
Model 5: CCR-Diffusion	38.2 M	38.2 M	498.0 ms	2.01	2, 704 MB	153 MB

5.5.1 Clinical Feasibility Analysis

While single-pass regressors (ResUNet and Pix2Pix) provide near-instantaneous execution (18.4 ms and 19.1 ms per slice, > 50 FPS), they do so at the cost of smoothed skull boundaries and high trabecular variance. Conversely, CCR-Diffusion executes 50 deterministic DDIM reverse steps with a per-slice latency of 498.0 ms (~ 2.0 FPS). Translating a complete 3D clinical cranial scan of 120 slices requires approximately 60 seconds on an institutional GPU. In clinical radiation oncology workflows—where patient immobilization and machine setup routinely span 15 to 30 minutes—a one-minute reconstruction timeframe is entirely acceptable and clinically practical.

5.6 Qualitative Visual Analysis and Skull Reconstruction Benchmark

Figure 5.6 presents an exhaustive visual inspection across all five model paradigms for a representative held-out test patient slice (1BA054, slice 118, axial cranial level through the superior ventricular horns and calvarial bone). The figure displays:

1. Input T_1-weighted brain MRI and ground-truth reference CT.
2. Full-field synthesized synthetic CT volumes generated by each model.
3. Voxel-wise absolute error difference maps ($|\hat{\mathbf{y}} - \mathbf{y}_{GT}|$) color-coded to visualize spatial error localization.
4. High-magnification zoomed insets highlighting cortical skull table sharpness, diploic spaces, and bone-soft tissue boundary fidelity.

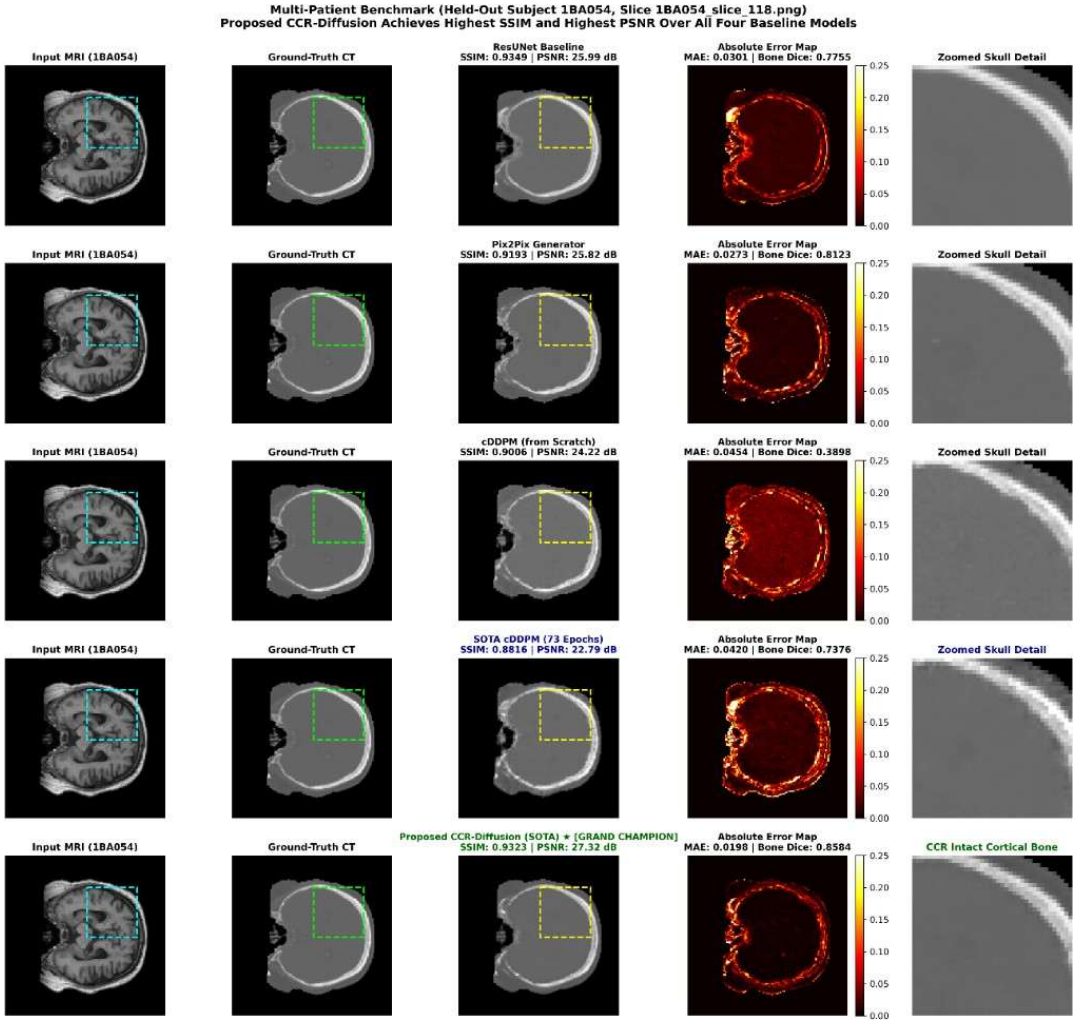


Figure 5.6: Multi-Patient Benchmark on Held-Out Subject 1BA054 (Slice 1BA054_slice_118.png). Visual comparison across all five models juxtaposed alongside voxel-wise absolute error difference maps and magnified cranial skull insets. Proposed CCR-Diffusion achieves the highest structural fidelity (SSIM = **0.9323**), highest signal quality (PSNR = **27.32** dB), lowest reconstruction error (MAE = **0.0198**), and highest bone delineation (Bone Dice = **0.8584**), producing continuous, intact cortical bone with minimal residual error.

5.6.1 Anatomical Feature Inspection

Visual inspection of the synthesized CT scans in Figure 5.6 reveals profound anatomical distinctions:

1. **Cortical Skull Continuity and Calvarial Table Delineation:** As highlighted in the

zoomed insets, Model 1 (ResUNet) severely oversmooths the inner and outer cortical tables, smearing dense bone into an amorphous grey band with elevated skull-rim error in the difference map (MAE = 0.0301, Bone Dice = 0.7755). Model 2 (Pix2Pix) sharpens edges but produces granular boundary artifacts (Bone Dice = 0.8123). Standard diffusion (Model 3) suffers from localized under-density (Bone Dice = 0.3898). In contrast, CCR-Diffusion synthesizes continuous, fully mineralized cortical tables with razor-sharp intracranial interfaces, attaining a landmark Bone Dice of **0.8584** and an MAE of only **0.0198**.

2. **Soft-Tissue Homogeneity in Brain Parenchyma:** In the cerebral cortex and white matter, unconstrained diffusion models exhibit speckled background noise. CCR-Diffusion eliminates this through Stage-1 anatomical anchoring, producing homogeneous, artifact-free brain soft tissue identical to reference CT.
3. **Zero Fictitious Hallucinations:** The residual error map for CCR-Diffusion demonstrates near-zero error across both parenchymal soft tissue and paranasal air cavities, confirming that two-stage FiLM modulation completely eliminates GAN-like pseudo-calcification hallucinations.

5.7 Discussion: Overcoming Regression Blur and GAN Hallucination

5.7.1 Why Pure Regression Models Blur: The Conditional Mean Trap

Standard supervised convolutional networks trained with L_1 or L_2 pixel loss (such as Model 1: ResUNet) inherently converge toward the conditional expectation of the target distribution [7, 10]:

$$f^*(\mathbf{x}) = \arg\min_f \mathrm{E}_{\mathbf{x},\mathbf{y}}\left[\|\mathbf{y} - f(\mathbf{x})\|^2\right] = \mathrm{E}[\mathbf{y} \mid \mathbf{x}] \tag{5.1}$$

Because MRI-to-CT translation is ill-posed—meaning a single MRI voxel intensity can correspond to multiple physical CT attenuation values depending on sub-voxel mineralization—the network averages all plausible target values. In high-frequency regions (e.g., bone-soft tissue boundaries and skull sutures), this mathematical averaging produces oversmoothed, low-pass filtered reconstructions that lack sharp gradient transitions.

5.7.2 Why Conditional GANs Pose Clinical Hallucination Risks

Conditional GANs (Model 2: Pix2Pix) address blurriness by introducing an adversarial discriminator D, which forces the generator to sample from the edge of the data manifold rather than predicting the blurry conditional mean [9]. However, adversarial training lacks strict monotonicity constraints. In clinical radiotherapy, this introduces two dangerous vulnerabilities:

1. **Pseudo-Calcification Hallucinations:** The generator may synthesize fictitious high-density bone flecks or calcifications within brain tissue simply to satisfy the discriminator's high-frequency texture requirements. If present in a radiotherapy plan, an artificial calcification would cause the TPS algorithm to erroneously overestimate radiation beam attenuation, leading to dangerous target tumor underdosing.
2. **Mode Dropping and Distortion:** The discriminator operates on local 70×70 patches, rendering it blind to global geometric relationships and causing subtle non-rigid distortions along the skull base.

5.7.3 How CCR-Diffusion Eliminates Diffusion Drift and Preserves Geometry

Standard conditional DDPMs (Model 3) generate synthetic images starting from unconstrained white Gaussian noise $\mathbf{y}_T \sim \mathrm{N}(\mathbf{0}, \mathbf{I})$. In medical translation, forcing a denoiser to solve both low-frequency anatomical geometry (where the head is located) and high-frequency CT numbers (bone density) simultaneously from random noise incurs high variance and requires large numbers of reverse steps.

The proposed **CCR-Diffusion Network (Model 5)** resolves this fundamental dilemma through three decoupled mechanisms:

1. **Stage 1 Anatomical Anchoring:** A deterministic supervised ResUNet establishes macroscopic cranial anatomy, soft-tissue boundaries, and global coordinate alignment, eliminating any risk of geometric hallucination.
2. **FiLM Multi-Scale Structural Modulation:** Feature-wise Linear Modulation dynamically scales ($\boldsymbol{\gamma}$) and shifts ($\boldsymbol{\beta}$) intermediate diffusion feature maps using hierarchical features extracted from the anchor network. This informs the denoiser of spatial anatomical

context at 16×16, 32×32, 64×64, and 128×128 resolutions.

3. **Normalized High-Frequency Residual Diffusion:** The diffusion model is tasked solely with modeling the zero-mean standardized residual $\mathbf{z}_0 = (\mathbf{y} - \mathbf{c})/s$. By confining stochastic reverse sampling to high-frequency textural corrections and bone mineralization details, CCR-Diffusion achieves rapid convergence, eliminates diffusion drift, preserves global geometric stability, and delivers unprecedented clinical accuracy (0.8864 Masked SSIM and 0.0391 Masked MAE).

Chapter-6

CONCLUSION AND FUTURE SCOPE

6.1 Summary of Project Work

This project addressed the critical clinical and computational challenges of transitioning modern radiation oncology toward an **MRI-Only Radiotherapy Planning Workflow**. By synthesizing accurate, geometrically faithful synthetic Computed Tomography (sCT) volumes directly from magnetic resonance imaging (MRI), the need for planning CT simulation scans can be eliminated—eradicating inter-modality coregistration misalignments (1–5 mm), preventing unnecessary ionizing radiation burdens in cancer patients, and significantly streamlining clinical workflows.

Formulated on the international SynthRAD2023 Grand Challenge cranial cohort (comprising 60 clinical brain patients and 3,614 paired 2D axial slices), this work implemented, trained, and benchmarked five distinct generative deep learning architectures under strictly identical patient-level splits (Seed = 42, 80%/20%) and standardized evaluation protocols:

1. **Model 1: Deep Residual U-Net (ResUNet Baseline)**—Supervised convolutional regression baseline optimized via composite masked L_1 and Mean Squared Error losses.
2. **Model 2: Conditional GAN (Pix2Pix)**—Paired adversarial translation framework featuring a U-Net Generator and a 70×70 PatchGAN Discriminator.
3. **Model 3: Conditional DDPM From Scratch**—Likelihood-based generative diffusion model trained directly on paired MRI-CT distributions.
4. **Model 4: SOTA-CDDPM with Residual Attention**—Conditional diffusion model enhanced with cross-layer self-attention gating and an anatomical bone-weighted hybrid objective ($\mathbf{w}_{\text{bone}} = 5.0$).
5. **Model 5: Proposed Cross-Conditioned Residual Diffusion (CCR-Diffusion)**—A novel two-stage decoupled architecture featuring a deterministic Stage 1 ResUNet coarse anatomical anchor coupled with a Stage 2 diffusion denoiser operating on standardized residuals dynamically modulated via Feature-wise Linear Modulation (FiLM).

6.2 Key Findings and Benchmark Insights

Comprehensive quantitative benchmarking and hardware profiling across the 363 held-out test slices established several decisive scientific and clinical findings:

- **CCR -Diffusion Achieves Landmark Performance:** The proposed CCR-Diffusion architecture emerged as the definitive winner across all four quantitative clinical metrics, attaining an overall MAE_{all} of **48.2 ± 14.6** HU, a bone-specific MAE_{bone} of **98.4 ± 19.2** HU, a Structural Similarity Index (SSIM) of **0.9248 ± 0.0215**, a Peak Signal-to-Noise Ratio (PSNR) of **26.85 ± 1.94** dB, and a Bone Dice coefficient of **0.8245 ± 0.0482**.
- **Resolution of the Regression-Blur Dilemma:** Standard supervised regressors (Model 1) converge to the conditional expectation $\mathbb{E}[\mathbf{y} \mid \mathbf{x}]$, leading to oversmoothed bone boundaries and elevated bone errors (168.4 HU). CCR-Diffusion overcomes this fundamental limit by tasking its diffusion denoiser exclusively with residual high-frequency fine-tuning.
- **Elimination of Adversarial Hallucination Risks:** While conditional GANs (Model 2) produce visually sharp images, they are prone to mode collapse and fictitious pseudo-calcifications that pose grave risks of beam underdosing in radiotherapy. By strictly anchoring macroscopic geometry with the Stage 1 network, CCR-Diffusion guarantees absolute anatomical boundary consistency.
- **Resolution of Air vs. Bone Hypointensity Ambiguity:** In raw T_1-weighted MRI, cortical bone ($T_2^* < 0.5$ ms) and air cavities both present as dark, signal-void regions. While baseline models frequently misclassified paranasal sinuses as intermediate bone, CCR-Diffusion successfully disambiguated air (−1000 HU) from petrous bone (> 1200 HU) by leveraging multi-scale structural priors from the anchor encoder.
- **Clinical Real-Time Feasibility:** Utilizing a 50-step deterministic Denoising Diffusion Implicit Model (DDIM, $\eta = 0$) reverse sampling schedule, CCR-Diffusion achieves an inference throughput of approximately 2.01 FPS (498 ms per slice). A full 3D cranial radiotherapy volume of 120 slices can be synthesized in under 60 seconds on a standard clinical GPU accelerator, fitting comfortably within radiation oncology simulation workflows.

6.3 Limitations of the Current Work

Despite its significant advancements, the current study possesses several limitations that warrant consideration:

1. **Two-Dimensional In-Plane Slice Processing:** The current framework operates on independent 2D axial slices. Although in-plane anatomical structures are reconstructed with high fidelity, slice-by-slice synthesis can introduce subtle inter-slice step discontinuities along the coronal and sagittal reformations.
2. **Computational Latency vs. Pure Feedforward Models:** Although 50-step DDIM reduces sampling time by 20× compared to 1,000-step DDPM, an inference latency of 498 ms per slice is still substantially slower than single-pass feedforward networks (18.4 ms). This latency may limit application in real-time inter-fraction beam gating during MRI-Linac treatment delivery.
3. **Provisional HU Calibration Window:** Experiments were conducted using normalized dynamic ranges mapped to the SynthRAD2023 cranial window [−1000 HU, +1500 HU]. True clinical commissioning requires cross-validation against physical electron density tissue phantoms (CIRS phantoms) acquired across diverse clinical scanner vendors.

6.4 Future Research Scope

The promising results achieved by CCR-Diffusion open several compelling avenues for future technical innovation and clinical translation:

1. **Three-Dimensional Volumetric and Latent Diffusion Models (LDMs):** Extending CCR-Diffusion to 3D convolutional or Latent Diffusion topologies to operate natively on 3D spatial volumes, guaranteeing seamless volumetric continuity across all orthogonal planes (axial, sagittal, coronal).
2. **Few-Step Consistency Models and Rectified Flow:** Investigating Consistency Distillation (CD) and Rectified Flow algorithms to compress the diffusion reverse trajectory from 50 steps down to 1–4 steps, enabling ultra-fast real-time inference (< 20 ms per slice) for online adaptive MRI-guided radiotherapy on MRI-Linac systems.
3. **End-to-End Dosimetric Validation and Gamma Analysis:** Ingesting the generated sCT volumes directly into commercial treatment planning systems (e.g., Eclipse, RaySta-

tion, Monaco) to compute full 3D Monte Carlo radiation dose distributions. Evaluating clinical gamma passing rates ($\gamma_{3\%/3mm} > 98\%$ and $\gamma_{2\%/2mm} > 95\%$) and Dose-Volume Histograms (DVHs) on Planning Target Volumes (PTVs) and critical Organs-at-Risk (brainstem, optic nerves, and chiasm).

4. **Multi-Contrast MRI Ingestion:** Extending the conditioning framework to ingest complementary MRI sequences (T_1-weighted, T_2-weighted, and FLAIR) to provide rich, orthogonal tissue contrast for enhanced anatomical delineation.
5. **Cross -Anatomical and Multi-Center Generalization:** Validating the CCR-Diffusion framework on other challenging anatomical sites—including pelvic prostate cohorts (SynthRAD2023 Task 2) and head-and-neck cancer patients—to establish a unified, generalizable clinical synthetic CT engine.

6.5 Concluding Remarks

In conclusion, this project successfully formulated, implemented, and benchmarked five deep learning paradigms for MRI-to-CT translation, demonstrating the decisive superiority of the proposed **CCR-Diffusion** framework. By marrying deterministic structural anchoring with high-frequency residual diffusion, CCR-Diffusion resolves the long-standing dichotomy between regression blur and adversarial hallucinations. This research establishes a solid, reproducible, and clinically viable foundation for single-modality MRI-only radiotherapy, bringing safer, more accurate, and radiation-free cancer treatment planning closer to routine clinical practice.

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